CASL-VAE: Learning Structured Latent Variables from Unpaired Data for Semi-supervised Clustering and Paired Sample Generation

Sai Spandana Chintapalli[1] **Pratik Chaudhari**[1,†] **Christos Davatzikos**[1,†]
[1] Center for AI and Data Science for Integrated Diagnostics (AI2D), University of Pennsylvania
{schinta, pratikac}@engineering.upenn.edu,
christos.davatzikos@pennmedicine.upenn.edu

Abstract

Quantifying variability in a target population relative to a reference population is central to many scientific and clinical problems (e.g., diseased vs. healthy). Yet, without paired data and in the presence of heterogeneous target variation, existing methods struggle to separate multiple modes of target-specific variation.We propose *CASL-VAE*, a deep contrastive latent variable model that learns structured latent generative factors from unpaired data. CASL-VAE factorizes variation into continuous common latent factors shared across populations and hierarchical salient latent factors that model target-specific heterogeneity as discrete subtypes and continuous within-subtype variation. Using variational inference, we show how approximate joint likelihood optimization over reference and target domains can be performed using unpaired data, providing a principled basis for paired-sample generation and cross-domain analysis. We validate CASL-VAE on semi-synthetic neuroimaging data, demonstrating improved subtype recovery and paired-sample generation compared to baseline clustering and generative models. We also validate its ability to reveal biologically plausible heterogeneity in Alzheimer's disease. Code is available at `https://github.com/SpandanaC-hub/CASLVAE`.

1 Introduction

Many scientific and applied problems involve understanding how a target population varies with respect to a reference population. For example, in medicine, researchers analyze pathological patterns in patient data relative to healthy controls [13]. Similar comparative analyses arise in genomics, drug discovery, and other biomedical settings, where contrasts between groups are often more informative than absolute measurements [34, 38, 4].

In practice, such comparisons are challenging for three reasons: (i) variation in the target domain is often entangled with variation present in the reference domain, obscuring subtle target-specific effects (for example, natural biological variation in healthy individuals could be mistaken for disease-related changes in patients); (ii) target-domain variation is frequently heterogeneous [32], reflecting latent subtypes or context-dependent factors (for instance, patients with the same diagnosis may have diverse underlying pathologies, or cell populations may respond differently to the same treatment); and (iii) paired observations across domains—where each target sample has a corresponding reference sample—are rarely available, preventing direct supervision for learning cross-domain relationships. As a result, existing methods often struggle to disentangle heterogeneous, target-specific variation in unpaired data [31, 14].

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Generative representation learning can help address these challenges by learning latent representations that model the data generation process [6]. In principle, such representations can differentiate between domains, capture heterogeneous subpopulations, and support downstream generative analyses [26]. However, standard generative models applied to multi-domain data typically entangle shared and domain-specific factors, leading to unstable or difficult-to-interpret representations [20, 8]. Contrastive Analysis (CA) methods aim to explicitly separate common variation shared across reference and target domains from salient variation unique to the target domain [2, 35]. Yet existing formulations often lack sufficient latent structure to capture multi-modal target heterogeneity or to preserve sample-level semantics across domains, limiting their ability to support downstream generative tasks such as paired data generation.

To address these limitations, we propose ***CASL-VAE*** *(Contrastive Analysis with Structured Latents via Variational Autoencoder)*, a generative framework for learning structured and disentangled latent representations from unpaired reference–target data (Figure 1). CASL-VAE factorizes the latent space into (i) a **continuous common latent** that captures variation shared across domains, and (ii) a **hierarchical salient latent** that models heterogeneous, target-specific variation via discrete latent subtypes and continuous within-subtype factors. The hierarchical design, regularized using domain-level labels (reference vs target labels)[‡], enables **unsupervised target subtype discovery**. This structured latent decomposition acts as an inductive bias that, together with sufficient constraints, enables approximate joint likelihood learning across domains using only marginal observations. As a result, CASL-VAE supports **paired-sample generation** and **interpretable cross-domain comparisons** despite the absence of paired training data.

We validate CASL-VAE on neuroimaging data, where brain measurements exhibit substantial inter-individual variability and heterogeneous pathological patterns are difficult to isolate. Moreover, paired measurements across conditions (e.g., the same participant observed in both healthy and diseased states) are generally unavailable, making unpaired contrastive modeling both practically relevant and technically challenging. While our experiments focus on neuroimaging-derived **regional brain volumes**, the proposed framework is broadly applicable to other settings involving unpaired reference–target comparisons. To demonstrate generalizability and clinical relevance, we evaluate CASL-VAE across multiple semi-synthetic and real-world datasets. Our **contributions** are threefold:

- We introduce CASL-VAE, a contrastive generative framework that learns disentangled shared and target-specific representations from unpaired data using structured latent variables.
- We propose a hierarchical salient latent decomposition that enables unsupervised discovery of target subtypes while preserving continuous within-subtype variation.
- We demonstrate that this structure enables interpretable downstream capabilities, including paired-sample generation and cross-domain comparison, and validate these properties empirically.

2 Related work

Probabilistic latent variable models such as variational autoencoders (VAEs) aim to learn compact representations of complex data [10], while conditional variants model data distributions conditioned on domain labels [37]. From a representation learning perspective, these models are primarily optimized for data generation. Without additional modeling assumptions or regularization [20], their latent representations may entangle shared and domain-specific sources of variation, limiting interpretability and downstream utility in comparative analyses.

Contrastive analysis (CA) addresses this by explicitly aiming to separate variation that is shared across two datasets from variation that is enriched in a target dataset. Early CA approaches relied on linear factor models [16, 2]. More recent work extends CA to nonlinear settings using probabilistic latent-variable models [1, 35], often with additional regularization to prevent information leakage and preserve the semantic meaning of the learned factors [39, 29]. While this enables separation of shared and target-specific variation, it does not explicitly model heterogeneous structure within the target population. In many real-world applications, however, separating two populations is not sufficient: the target population itself often exhibits meaningful heterogeneity that is critical to understand. For example, CA has been applied to single-cell RNA-seq [40, 24] and neuroimaging [3, 44] datasets, where target-specific variation is frequently multi-modal and reflects distinct biological subtypes or phenotypes that need to be identified. In these settings, heterogeneity is typically addressed post

[‡]In this paper, the term *semi-supervised* refers to supervision available only at the domain level (reference vs. target), without any annotations for target subtypes or paired samples.

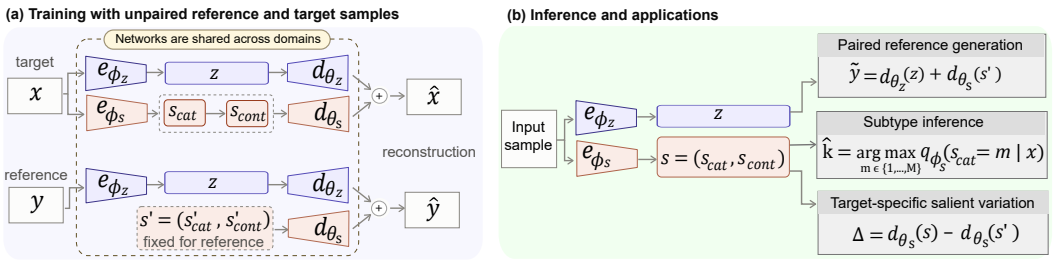


Figure 1: (a) CASL-VAE is trained on unpaired target (x) and reference (y) samples, with the common encoder e_{ϕ_z} learning variation shared across domains through common latent z, and the salient encoder e_{ϕ_s} learning target-specific variation. The salient latent has a hierarchical structure comprising a discrete subtype variable $s_{\text{cat}} \in \{1, \ldots, M\}$ and a continuous within-subtype variable s_{cont}. For reference samples, salient latents are constrained to uninformative preset values $s' = (s'_{\text{cat}}{=}1,\ s'_{\text{cont}}{=}0)$. So category 1 is reserved for the reference domain and the remaining categories $2, \ldots, M$ encode target subtypes. The common and salient decoder outputs are summed to reconstruct each input. (b) CASL-VAE support a range of downstream applications; examples include subtype inference, paired reference generation, and quantification of target-specific salient variation.

hoc, for example, by clustering the learned salient representations. However, when target effects are subtle or subtypes are partially entangled, such representations may not form well-separated clusters. This issue is further exacerbated in VAE-based CA models, which can be prone to posterior collapse, potentially conflating or collapsing multi-modal target variation unless additional structure is imposed.

Clustering-based approaches have also been proposed for capturing target heterogeneity [15]. Standard clustering methods, however, do not leverage contrastive structure and therefore may produce clusters that are driven by shared variation rather than target-specific effects. Some recent methods cluster based on differences between target and reference populations [7, 42]. While effective for unsupervised subtype discovery, these approaches typically optimize discrete clustering objectives rather than learning reusable generative latent representations and therefore have limited interpretability.

CASL-VAE addresses these limitations by integrating contrastive analysis with structured hierarchical latent variables within a single generative framework. Designed for unpaired reference–target data, it enables the modeling of target subtypes and within-subtype variability. The framework supports data-driven, unsupervised discovery of target-specific heterogeneity while remaining interpretable, distinguishing it from many prior contrastive and clustering-based approaches that rely on post-hoc analyses or subtype labels.

3 CASL-VAE

3.1 Problem Setup

Given unpaired observations from two related domains, a target domain $\mathcal{X} = \{x_i\}_{i=1}^{N_x}$, $x_i \sim p(x)$ and a reference domain $\mathcal{Y} = \{y_j\}_{j=1}^{N_y}$, $y_j \sim p(y)$ where no correspondence is assumed between sample i and j. Our goal is to learn latent representation that separates variation shared across these domains from heterogeneous, multi-modal variation specific to the target domain, enabling target-subtype discovery and cross-domain comparisons.

3.2 Contrastive Latent Variables as Generative Factors

We introduce three latent variables: a continuous common latent variable $z \in \mathbb{R}^k$, a discrete (categorical) salient latent variable $s_{\text{cat}} \in \{1, ..., M\}$, and a continuous salient latent variable $s_{\text{cont}} \in \mathbb{R}^t$ conditioned on s_{cat}. The common latent models variation shared across both domains and is assumed to fully encode the variation present in the reference domain. The salient latent variables provide a **hierarchical representation** of target-specific variation, where the discrete component captures coarse subtype structure and the continuous component models finer-grained variation within

these subtypes (Appendix Figure 5). So, the *prior* factorizes as

$$p(z, s_{\text{cat}}, s_{\text{cont}}) = p(z)p(s_{\text{cat}}, s_{\text{cont}}) = p(z)\, p(s_{\text{cat}})\, p(s_{\text{cont}} \mid s_{\text{cat}}), \tag{1}$$

where $p(z) = \mathcal{N}(0, I)$, $p(s_{\text{cat}}) = \text{Uniform}\{1, \ldots, M\}$, and $p(s_{\text{cont}} \mid s_{\text{cat}}) = \mathcal{N}(\mu_\psi(s_{\text{cat}}), I).\mu_\psi(s_{\text{cat}})$ is a learnable embedding function that maps each category s_{cat} to a subtype-specific prior mean.
Remark: In neuroimaging applications, z may encode demographic or scanner variability shared between healthy and diseased participants, while s_{cat} and s_{cont} may encode disease-specific pathological patterns of variability.

3.3 Optimizing the Paired Data Likelihood from Unpaired Data

Given z and $s = (s_{\text{cat}}, s_{\text{cont}})$, the target and reference observations are conditionally independent, i.e., $p(x, y \mid z, s) = p(x \mid z, s)p(y \mid z)$. Similarly, the corresponding variational posterior factorizes as $q(z, s \mid x, y) = q(z \mid x, y)q(s \mid x)$. This leads to a variational lower bound on the joint log-likelihood $\log p(x, y)$ which depends on the posterior term $q(z|x, y)$ (Appendix A.1, Eq.18), and therefore requires paired samples. Since paired samples are unavailable, this term cannot be directly inferred.

To enable learning the joint likelihood from unpaired data, we introduce *posterior decoupling*,

$$q(z|x, y) \approx q(z|x) \approx q(z|y), \tag{2}$$

which enforces a domain-invariant common latent representation that can be inferred from either domain x or y. In practice, this constraint is implemented via regularization (Section 3.6).

Under the posterior decoupling assumption, the variational lower bound simplifies to

$$\begin{aligned}\log\ p(x, y) \geq\ & \mathbb{E}_{q(z|x)q(s|x)} \log p(x|z, s) + \mathbb{E}_{q(z|y)} \log p(y|z) - \tfrac{1}{2}\text{KL}(q(z|x)||p(z)) \\ & - \tfrac{1}{2}\text{KL}(q(z|y)||p(z)) - \text{KL}(q(s|x)||p(s)),\end{aligned} \tag{3}$$

where the KL regularization on the common latent is evenly split across domains to preserve symmetry and avoid over-regularization. The full derivation is provided in Appendix A.1.

Thus, the paired-data variational objective decomposes into separate domain-specific terms that can be optimized using only unpaired observations. The resulting objective is equivalent in form to those commonly used in contrastive analysis [1, 39, 29]. This provides a principled probabilistic justification for the common-salient latent decomposition, and explains why contrastive objectives can be viewed as implicitly maximizing a relaxed joint likelihood from unpaired data, establishing a theoretical bridge for paired-sample generation and cross-domain analysis.

3.4 Parameterization of the Latent Inference Model and the Generative Model

The inference model consists of two neural networks: a common encoder network e_{ϕ_z} and a salient encoder network e_{ϕ_s}, which together define a factorized variational posterior over the latent variables. For a target domain sample $x \in \mathcal{X}$, the variational *posterior* factorizes as

$$q_\phi(z, s_{\text{cat}}, s_{\text{cont}} \mid x) = q_{\phi_z}(z \mid x)\, q_{\phi_s}(s_{\text{cat}}, s_{\text{cont}} \mid x) = q_{\phi_z}(z \mid x) q_{\phi_s}(s_{\text{cont}} \mid s_{\text{cat}}, x) q_{\phi_s}(s_{\text{cat}} \mid x) \tag{4}$$

The individual variational factors are parameterized as $q_{\phi_z}(z \mid x) = \mathcal{N}\big(z \mid \mu_z(x), \sigma_z^2(x)I\big)$, $q_{\phi_s}(s_{\text{cat}} \mid x) = \text{Gumbel-Softmax}(\alpha(x), \tau)$, and $q_{\phi_s}(s_{\text{cont}} \mid s_{\text{cat}},\ x) = \mathcal{N}(s_{\text{cont}} \mid \mu_s(s_{\text{cat}},\ x), I)$. Here, common encoder e_{ϕ_z} outputs $\mu_z(x)$ and $\sigma_z(x)$, while salient encoder e_{ϕ_s} outputs the logits $\alpha(x)$ and the conditional mean $\mu_s(s_{\text{cat}}, x)$. The discrete latent is modeled using a Gumbel–Softmax distribution, where the temperature τ controls the relaxation of the categorical variable [23]. During training, s_{cat} lies on the probability simplex Δ^{M-1}, enabling gradient-based optimization. At test time, it is discretized to a hard one-hot vector.

Note: For reference-domain samples $y \in \mathcal{Y}$, only $q_{\phi_z}(z \mid y)$ is inferred; the salient latents are fixed to an uninformative preset state $s' = (s'_{\text{cat}},\ s'_{\text{cont}})$ with $s'_{\text{cat}} = 1$ (reference category) and $s'_{\text{cont}} = 0$. This constraint prevents the salient latent from encoding reference-domain variation and encourages it to capture target-specific variability (Section 3.6).

For the generative model, we parameterize the relation between the latent variables and the observed data with two neural networks: a common decoder d_{θ_z} and a salient decoder d_{θ_s}, giving

$$\begin{aligned}p_\theta(x \mid z, s_{\text{cat}}, s_{\text{cont}}) &= \mathcal{N}\big(x \mid d_{\theta_z}(z) + d_{\theta_s}(s_{\text{cat}}, s_{\text{cont}}), I\big), \\ p_\theta(y \mid z) &= \mathcal{N}\big(y \mid d_{\theta_z}(z) + d_{\theta_s}(s'_{\text{cat}}, s'_{\text{cont}}), I\big).\end{aligned} \tag{5}$$

The additive decoder structure provides interpretability by explicitly separating common and salient contributions in the input space. For reference samples, the salient latent is fixed to $s' = (s'_{\text{cat}}, s'_{\text{cont}})$; thus, $d_{\theta_s}(s')$ acts as a constant baseline offset in the reference reconstruction.

This modeling allows direct interpretation and visualization of **prototypical target patterns of variation** as

$$\delta_{s_{\text{cat}}} = d_{\theta_s}(s_{\text{cat}}, \mu_\psi(s_{\text{cat}})) - d_{\theta_s}(s'_{\text{cat}} = 1, s'_{\text{cont}} = 0), \tag{6}$$

for each $s_{\text{cat}} \in \{2, \ldots, M\}$. This highlights the target-specific subtype patterns relative to the reference domain.

Note: We use a Gaussian likelihood for the observations because our application involves continuous brain volumes. However, the model can be adapted to other likelihoods (e.g., Bernoulli) for different data types.

3.5 Evidence Lower Bound (ELBO) for CASL-VAE

Expanding eq. 3 (see Appendix A.2 for details), the ELBO for CASL-VAE is

$$\mathcal{L}_{ELBO} = \mathcal{L}_x + \mathcal{L}_y, \tag{7}$$

where

$$\begin{aligned}\mathcal{L}_x = {} & \mathbb{E}_{q_{\phi_z}(z|x) q_{\phi_s}(s_{\text{cat}}, s_{\text{cont}}|x)} \log p(x \mid z, s_{\text{cat}}, s_{\text{cont}}) - \frac{1}{2} KL(q_{\phi_z}(z \mid x) || p(z)) \\ & - KL(q_{\phi_s}(s_{\text{cat}} \mid x) || p(s_{\text{cat}})) - \mathbb{E}_{q_{\phi_s}(s_{\text{cat}}|x)}[KL(q_{\phi_s}(s_{\text{cont}} \mid s_{\text{cat}}, x) || p(s_{\text{cont}} \mid s_{\text{cat}}))],\end{aligned} \tag{8}$$

$$\mathcal{L}_y = \mathbb{E}_{q_{\phi_z}(z|y)} \log p(y \mid z) - \frac{1}{2} KL(q_{\phi_z}(z \mid y) || p(z)). \tag{9}$$

This decomposition corresponds to the marginal ELBOs for the target and reference domains under the hierarchical latent structure.

3.6 Regularizations for structured common-salient latent decomposition

To encourage a domain-invariant posterior $q_{\phi_z}(z \mid x) \approx q_{\phi_z}(z \mid y)$, we apply two complementary regularizers: a global distribution alignment term (MMD) and, under cross-domain transformations, a sample-level consistency term that preserves sample identity. We introduce two additional regularization terms that jointly suppress encoding of reference-domain variability within the salient latent space. Together, these terms provide complementary inductive biases that support the structured common–salient decomposition and discourage degenerate latent solutions(Appendix A.5).

Maximum Mean Discrepancy (MMD) term encourages the common latent distributions inferred from the two domains to be aligned [17], Appendix A.3:

$$\mathcal{R}_{\text{MMD}} = \text{MMD}^2(q_{\phi_z}(z \mid x),\, q_{\phi_z}(z \mid y))\,. \tag{10}$$

Sample-level Latent Consistency term enforces that the common latent representation is stable under cross-domain transformation from $\mathcal{X}$ to $\mathcal{Y}$, which provides sample-level latent alignment $q_{\phi_z}(z \mid x) \approx q_{\phi_z}(z \mid \tilde{y})$ [45]:

$$\mathcal{R}_{\text{LC}} = \mathbb{E}_{x \sim p(x),\, z \sim q_{\phi_z}(z|x)} \left\| \mu_z(\tilde{y}) - \mu_z(x) \right\|_1, \tag{11}$$

$$\text{where} \quad \tilde{y} = d_{\theta_z}(z) + d_{\theta_s}(s'_{\text{cat}}, s'_{\text{cont}}). \tag{12}$$

Here, $\tilde{y}$ is a conditionally generated paired reference sample obtained by decoding z with the salient latent fixed to the uninformative preset values.

Semi-supervised Binary Cross-Entropy term uses domain labels (reference vs. target) as weak supervision to prevent the discrete salient latent from capturing reference-domain structure. Reference samples are encouraged to map to preset category $s'_{\text{cat}} = 1$, while target samples are discouraged from this category, leaving the remaining $M - 1$ categories for unsupervised target subtype discovery:

$$\mathcal{R}_{\text{BCE}} = -\mathbb{E}_{y \sim p(y)} \log q_{\phi_s}(s_{\text{cat}} = 1 \mid y) - \mathbb{E}_{x \sim p(x)} \log\big(1 - q_{\phi_s}(s_{\text{cat}} = 1 \mid x)\big). \tag{13}$$

Reference Regularization of the Continuous Salient Latent. Although $\mathcal{R}_{\text{BCE}}$ separates reference and target samples in the discrete latent space, the continuous latent s_{cont} could still encode reference-domain variation. To prevent this, we regularize both the posterior and learned prior of s_{cont} for

reference samples:

$$\begin{aligned}\mathcal{R}_{\text{ref}} &= \mathrm{KL}(q_{\phi_s}(s_{\text{cont}} \mid y, s'_{\text{cat}}) \,\|\, \mathcal{N}(0, I)) + \mathrm{KL}(p(s_{\text{cont}} \mid s'_{\text{cat}}) \,\|\, \mathcal{N}(0, I)) \,, \\ &= \tfrac{1}{2} \left\| \mu_s(y, s'_{\text{cat}}) \right\|_2^2 + \tfrac{1}{2} \left\| \mu_\psi(s'_{\text{cat}}) \right\|_2^2. \end{aligned} \tag{14}$$

where the final expression follows from both distributions having identity covariance. This penalty drives both posterior and prior means toward zero, making s_{cont} uninformative for reference samples. *Information-Theoretic Interpretation:* $\mathcal{R}_{\text{BCE}}$ and $\mathcal{R}_{\text{ref}}$ jointly minimize an upper bound on the mutual information between reference domain $\mathcal{Y}$ and the salient latent $(s_{\text{cat}}, s_{\text{cont}})$ (Appendix A.4). The former collapses the discrete salient latent for $\mathcal{Y}$, while the latter suppresses information in the continuous latent, ensuring that the salient latent variables capture only target-specific variation.

3.7 Overall Training Objective

The final training objective minimizes the negative ELBO along with the additional regularizers:

$$\mathcal{J} = -\mathcal{L}_{ELBO} + \lambda \mathcal{R}_{\text{MMD}} + \eta \mathcal{R}_{\text{LC}} + \zeta \mathcal{R}_{\text{BCE}} + \gamma \mathcal{R}_{\text{ref}}, \tag{15}$$

where λ, η, ζ, and γ control the strength of each regularizer. All expectations are approximated via Monte Carlo sampling with the reparameterization trick.

3.8 Inference and Generation

Besides providing a rich latent representation of reference-target data, the trained model supports three inference/generation modes:

1. **Subtype inference:** For each sample, the salient encoder outputs logits $\alpha(\cdot)$ over the M categories. After softmax normalization, samples are assigned to the category with highest probability. Category 1 represents reference domain, while the remaining categories represent target subtypes.
2. **Conditional paired-sample generation:** Given a sample, a corresponding cross-domain sample is obtained by decoding its inferred common latent representation under different salient states. Specifically, the common encoder infers $z \sim q_{\phi_z}(z \mid \cdot)$. Decoding z with the preset state $s' = (s'_{\text{cat}}, s'_{\text{cont}})$ produces a paired reference-domain sample $\tilde{y}$, while decoding with alternative salient states produces paired target-domain samples $\tilde{x}$.
3. **Synthetic data generation:** Synthetic samples are generated by sampling from the model priors, i.e., $z \sim p(z)$, $s_{\text{cat}} \sim p(s_{\text{cat}})$ and $s_{\text{cont}} \sim p(s_{\text{cont}} \mid s_{\text{cat}})$. Paired synthetic samples reuse the same z across domains, while unpaired samples draw independent z for each domain.

4 Parsing Neuroanatomical Heterogeneity in Disease Cohorts

Characterizing unlabeled disease heterogeneity despite confounding biological variability is a central challenge in neuroimaging studies of brain disorders. Such analyses can improve our understanding of disease mechanisms and support personalized diagnosis and treatment planning. Here, we apply CASL-VAE to semi-synthetic and real disease cohorts to evaluate (i) target-subtype discovery, (ii) paired-sample generation, and (iii) clinically meaningful characterization of disease heterogeneity.

4.1 Experimental Setup

Across all experiments, models are trained on unpaired reference-target samples using a fixed architecture with common latent dimension $k = 15$, discrete salient latent dimension M (selected based on clustering stability), and continuous salient latent dimension $t = 2$ for direct 2-D visualization of within-subtype variation. See Appendix A.6 for implementation details.

Data Preparation. We use T1-weighted Magnetic Resonance Imaging (T1w MRI) and clinical measurements from the iSTAGING consortium [19], which integrates multiple aging and neurodegenerative disease cohorts. From baseline scans, 145 regional brain volumes were extracted using a multi-atlas label fusion method [12], harmonized to remove site effects [33], and used as input features for CASL-VAE.

Semi-synthetic cohort: For controlled validation with ground-truth disease patterns and severity, we constructed semi-synthetic datasets using cognitively normal (CN) participants as the reference group

Table 1: Clustering performance for semi-synthetic data under mild (10–20%) and typical (10–30%) atrophy rates. Mean ARI with 95% confidence interval (CI) over 10 repeated cross-validation analysis. ARI ranges from 0 (random clustering) to 1 (perfect clustering). **Bold**: best.

Model	ARI ↑ (target-only)		ARI ↑ (target+reference)	
	10–20%	10–30%	10–20%	10–30%
GMM	0.001 (−0.005, 0.007)	0.232 (0.070, 0.394)	0.001 (−0.006, 0.008)	0.043 (0.009, 0.076)
GMVAE	0.038 (0.006, 0.069)	0.284 (0.129, 0.439)	–	–
K-means	−0.002 (−0.009, 0.004)	0.533 (0.423, 0.643)	0.006 (0.001, 0.012)	0.112 (0.093, 0.130)
cVAE	0.183 (0.124, 0.242)	0.654 (0.581, 0.726)	0.009 (0.000, 0.019)	0.041 (0.018, 0.064)
MM-cVAE	0.258 (0.150, 0.365)	0.320 (0.230, 0.411)	0.190 (0.142, 0.237)	0.428 (0.369, 0.486)
SepVAE	0.251 (0.196, 0.306)	0.611 (0.558, 0.664)	0.159 (0.130, 0.188)	0.451 (0.380, 0.523)
Smile-GAN	0.183 (0.097, 0.293)	**0.657 (0.508, 0.750)**	–	–
CASL-VAE	**0.468 (0.371, 0.564)**	0.620 (0.535, 0.706) *	**0.587 (0.521, 0.653)**	**0.697 (0.672, 0.722)**

Note: * ARI (target-only) of CASL-VAE is not significantly different from best (CIs overlap).

and a target group generated by simulating regional brain atrophy in an independent CN cohort. The target group is partitioned into three latent subtypes (S1–S3), each defined by volumetric reductions in distinct brain regions (Appendix Table 7). This preserves realistic inter-individual variability while enabling controlled simulation of disease effects at predefined atrophy rates.

Real disease cohort: For real-world clinical application, we used the ADNI dataset [5]. CN individuals served as the reference domain, while individuals with mild cognitive impairment (MCI) and Alzheimer's Disease (AD) formed the target domain.

Appendix B.1 provides full dataset details, including preprocessing, cohort construction, and train/test splits.

Evaluation metrics and baselines. Subtype discovery (clustering performance) is evaluated using Adjusted Rand Index (ARI) [22]. Paired-sample generation is assessed via cosine similarity between generated and ground-truth samples, and reconstruction quality via mean absolute error (MAE). We compare CASL-VAE against several unsupervised and weakly-supervised methods: (i) standard clustering (K-means, Gaussian mixture modeling (GMM)), (ii) deep clustering (GMVAE [36]), (iii) semi-supervised clustering (Smile-GAN [42]), (iv) conditional generative modeling (cVAE [11]), and (v) representative VAE-based CA models (SepVAE [29], MM-cVAE [40]). To evaluate clustering performance of cVAE, SepVAE, and MM-cVAE, we apply K-means to their posterior means (unified latent for cVAE; salient latent for CA models). See Appendix B.2 for details.

Selecting the optimal dimension of s_{cat} (M). In real neuroimaging data, the number of target subtypes is unknown; therefore, M is selected based on clustering stability under repeated hold-out cross-validation. For each candidate $M \in \{2, 3, 4, 5, 6\}$, we perform 10 repetitions of hold-out CV: in each repetition, the model is trained on 80% of the training data, and subtype assignments are inferred for the full training set. We then compute ARI between all pairs of clusterings across repetitions, and use the mean pairwise ARI as a stability score for each M. The value of M with the highest stability score is selected.

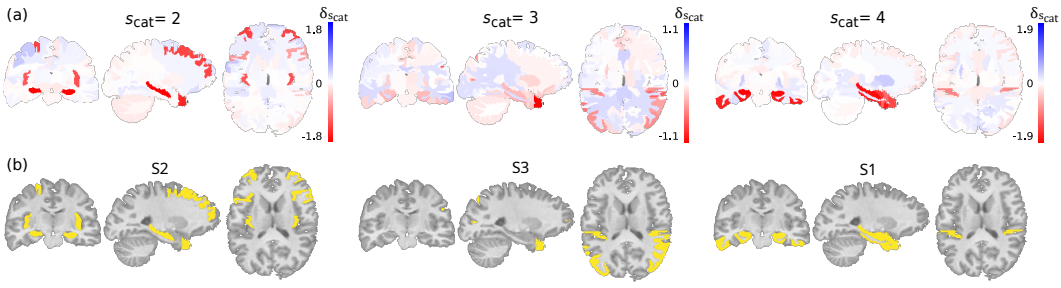


Figure 2: Results from semi-synthetic neuroimaging experiment with 10–30% atrophy. a) Prototypical target patterns $\delta_{s_{\text{cat}}}$ learned by CASL-VAE for $s_{\text{cat}} = 2, 3, 4$. Negative values (red) indicate volumetric reductions with respect to reference. b) Ground-truth subtype (S2,S3,S1) patterns; yellow = binary mask.

4.2 Results from Semi-synthetic Cohort Analyses

Subtype discovery. We first validate the stability-based selection of the categorical salient dimension M. Across semi-synthetic datasets with recoverable target-specific variation, clustering stability correctly selects $M = 4$, matching the ground-truth number of categories (Appendix B.3). With M selected, we evaluate both target-only and joint (target + reference) clustering. Target-only clustering assesses recovery of the three target subtypes, whereas joint clustering additionally assesses whether reference samples are correctly distinguished from the target subtypes. Table 1 reports clustering performance under two atrophy regimes: a typical regime (10–30%) corresponding to standard simulated effect sizes, and a mild regime (10–20%) highlighting a more challenging setting with weaker effect sizes.

Under the typical atrophy setting, CASL-VAE's performance is on par with the best baseline (SmileGAN), suggesting that multiple methods can recover subtype structure when disease effects are pronounced. Importantly, CASL-VAE provides an interpretable representation of the learned salient latent structure. Figure 2 shows brain maps of the prototypical target patterns learned by the model alongside the ground-truth subtype masks used in simulation. We observe near-perfect correspondence between the inferred patterns and the ground-truth masks, providing visual evidence that CASL-VAE recovers meaningful and interpretable subtype factors. In addition to recovering subtype-level patterns, CASL-VAE also captures within-subtype variability through the continuous salient latent space, as shown in Figure 3.

In the mild atrophy setting, CASL-VAE significantly outperforms all baselines in target-only clustering. It also achieves the highest ARI in the joint (target + reference) clustering setting in both the typical and mild regimes, demonstrating that it can simultaneously discover target subtypes and distinguish them from the reference domain. These gains can be attributed to CASL-VAE's contrastive latent structure and hierarchical modeling of the salient space. Importantly, this structured latent factorization does not compromise its reconstruction quality (Appendix Tables 14 and 15).

Conditional paired-sample generation. We evaluated paired-sample generation on semi-synthetic test datasets with paired reference and target samples. For each target sample, we generate a paired reference $\tilde{y}$ (Eq. 12) and measure its cosine similarity with the ground-truth paired reference sample, assessing their semantic similarity. CASL-VAE achieved the highest mean cosine similarity under both mild and typical atrophy rates (Table 2). Under the typical atrophy regime, CASL-VAE significantly outperformed all baselines. Under the mild atrophy regime, CASL-VAE significantly outperformed cVAE and SepVAE, while no significant difference was observed relative to MM-cVAE.

Ablation studies. We evaluate robustness to target-effect size, regularization strengths, and latent dimensionality (Appendix B.4.1-B.4.2).Subtype recovery deteriorates when target effects are extremely weak (0–10% atrophy). Performance is largely insensitive to the dimensionality of the continuous salient latent space, whereas smaller common latent dimensions increase reconstruction error. Notably, performance is strongest when all regularizers are active and remains stable across a range of hyperparameter values, indicating that the regularizers address complementary failure modes while avoiding excessive sensitivity to tuning. We further evaluated posterior decoupling by comparing the paired and decoupled variational objectives (Eq. 18 *vs* Eq. 3) across the simulated atrophy regimes (Appendix B.4.3). Although the ELBO gap increased with effect size, the relative

Table 2: Paired sample generation (target → reference) under mild (10–20%) and typical (10–30%) atrophy. Values are mean cosine similarity between generated and ground-truth references (higher is better), with 95% CI across 10 folds. † Significant vs. cVAE and SepVAE; * significant vs. all baselines (one-sided paired Wilcoxon signed-rank test over folds, Bonferroni-corrected α = 0.05/3 = 0.0167).

Model	10–20%	10–30%
cVAE	0.5471 (0.5428, 0.5514)	0.5208 (0.5170, 0.5247)
SepVAE	0.5287 (0.5188, 0.5387)	0.5409 (0.5326, 0.5493)
MM-cVAE	0.5780 (0.5728, 0.5832)	0.5666 (0.5578, 0.5755)
CASL-VAE	0.5835 (0.5777, 0.5894)†	0.5816 (0.5747, 0.5885)*

Table 3: Correlation between ADNI clinical variables and the overall deviation index (DI). * denotes $p < 0.001$.

Variable	Correlation	
ADNI-MEM	-0.469	*
ADNI-LAN	-0.348	*
ADNI-EF	-0.276	*
ADAS-COG-13	0.473	*
APOE E4 Carrier	0.242	*
MMSE	-0.354	*
CSF-Abeta	-0.254	*
CSF-PTau	0.146	
CSF-Tau	0.115	

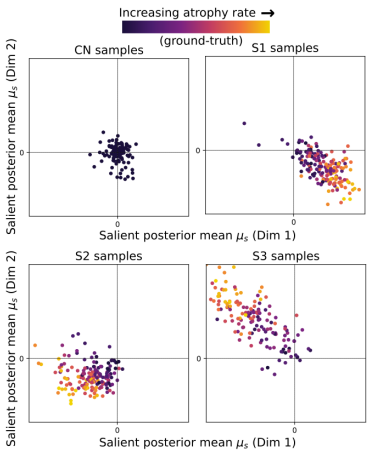


Figure 3: Conditional continuous salient 2-D latent means from held-out CN and target samples (10-30% simulated atrophy). Each point represents a sample's posterior mean $\mu_s(\cdot)$. The cross-hair marks the preset value s'_{cont}. Color indicates the ground-truth atrophy rate.

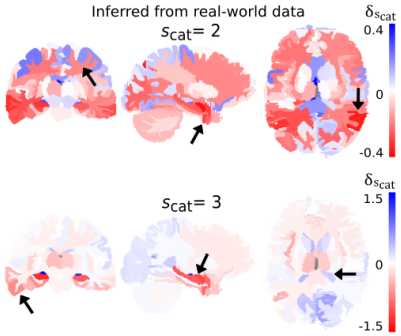


Figure 4: Prototypical target patterns $\delta_{s_{\text{cat}}}$ learned in ADNI MCI and AD participants relative to CN participants; arrows point to some clinically relevant regions. Negative values (red) indicate volumetric reductions with respect to CN group.

gap remained below 0.6% across all regimes, suggesting only a minor discrepancy between the paired and decoupled formulations.

4.3 Results from Real Disease Cohort Analysis

Subtype discovery. In the ADNI dataset, CASL-VAE achieves the highest clustering stability when the discrete salient latent has dimension M=3 (Appendix Figure 7). Figure 4 illustrates the resulting prototypical target-subtype patterns. The first pattern (top, $\delta_{s_{\text{cat}}=2}$) is characterized by relatively mild, diffuse brain atrophy involving regions such as frontal lobe, entorhinal cortex, and angular gyrus. In contrast, the second pattern (bottom, $\delta_{s_{\text{cat}}=3}$) shows more pronounced volumetric reductions concentrated in the medial temporal lobe (e.g. temporal gyrus and hippocampus), consistent with established signatures of AD pathology [9].

The learned categories also differed in diagnostic composition and MCI progression status (Appendix Table 16). Although Category 3 was enriched for AD participants and MCI converters, diagnostic overlap across categories and differences in regional atrophy patterns suggest that the learned subtype partition is not solely explained by disease stage.

Individual-level inference. We further validated the clinical utility of CASL-VAE by quantifying mean absolute target-specific salient variation for each individual in the held-out ADNI1 dataset. This acts as a deviation index (DI), quantifying each participant's overall deviation relative to the reference domain (Appendix B.6.2). The DI showed moderate to strong associations with ADNI clinical and biomarker variables (Table 3), suggesting that the learned salient representations reflect clinically relevant neuroanatomical variability associated with disease pathology.

5 Generalization to Other Datasets and Modalities

We evaluate CASL-VAE on three heterogeneous datasets spanning different modalities (Appendix B.7) to assess the generalizability of the proposed framework across diverse data settings. These datasets differ in dimensionality, measurement type, and latent structure.

Genomics (cell infection response): We apply CASL-VAE to gene expression data from mouse intestinal epithelial cells [18]. The reference group consists of uninfected cells, while the target group includes cells with two biologically distinct infection states (Salmonella and Heligmosomoides polygyrus). CASL-VAE recovers both infection-specific subtypes while separating them from the reference population, outperforming baseline CA models in clustering accuracy (Appendix Table 17,

Figure 8).

Multi-study neuroimaging cohort: We evaluate CASL-VAE on an independent cohort of cognitively normal and impaired participants [30, 19]. The model identifies two stable target subtypes that are associated with clinically relevant biomarker variation (Appendix Table 18, Figure 9), demonstrating the model's ability to learn structured common-salient representations across neuroimaging cohorts.

Toy image dataset: To test whether CASL-VAE captures heterogeneity in non-biological data, we construct a synthetic dataset using MNIST [43], where digit 0 serves as the reference class while target samples are constructed by superimposing digits 1, 2, or 3 onto digit 0 images. CASL-VAE correctly identifies salient subtype and within-subtype structure related to digits 1-3 while capturing shared variability related to digit 0 in the common latent (Appendix Figure 10).

Across all datasets, we use the same model configuration as in Section 4, without dataset-specific hyperparameter tuning. Despite substantial differences in domain characteristics, CASL-VAE consistently learns meaningful common and hierarchical salient latent decomposition. These results support the robustness of the proposed framework across diverse data settings.

6 Discussion

We introduced CASL-VAE, a structured contrastive latent-variable framework for learning shared and target-specific variation from unpaired reference-target data. By combining a hierarchical salient representation with an unpaired contrastive objective, CASL-VAE enables unsupervised subtype discovery, characterization of within-subtype heterogeneity, and paired-sample generation, as demonstrated empirically across diverse datasets.

Biological interpretation and validation. CASL-VAE learns structured target-specific variation without imposing biological priors or predefined disease progression trajectories. Consequently, the learned salient categories need not correspond uniquely to biological disease subtypes and may instead reflect disease severity, progression, or combinations of underlying pathological processes. Although the ADNI categories exhibit distinct anatomical patterns, these observations alone do not establish biologically distinct disease subtypes. Establishing biological validity therefore requires post-hoc evaluation using independent clinical, molecular, or longitudinal evidence.

Limitations. Like other CA methods, the latent factors learned by CASL-VAE are not identifiable without additional assumptions [27]. Although the proposed regularization terms constrain the solution space and empirically encourage meaningful common-salient decomposition, multiple equivalent factorizations may exist. In addition, CASL-VAE assumes that target heterogeneity can be represented using discrete salient categories. When heterogeneity is predominantly continuous or reflects overlapping latent mechanisms, enforcing discrete assignments may lead to unstable splitting or merging of latent categories. Sensitivity analyses further show that severe subtype imbalance, reference-cohort contamination, and misspecification of the category number M can degrade subtype recovery and distort the learned salient structure (Appendices B.4.4-B.4.6). Finally, CASL-VAE does not explicitly model covariate information, such as demographic or clinical variables. Their effects may therefore be encoded into either the common or salient latent space depending on whether they contribute to shared or target-specific variation. Extending the framework to incorporate such covariates through conditioning or supervision is an important direction for future work.

Broader impact. CASL-VAE addresses a key bottleneck in high-dimensional biomedical data analysis, where reference-target populations present strong inter-individual variability, unlabeled heterogeneous target structure, and a lack of paired observations, challenging direct group comparisons. By learning interpretable, structured contrastive latent representations from unpaired data, CASL-VAE enables both subtype- and sample-level characterization of target heterogeneity, providing a framework for more fine-grained disease stratification and exploratory analysis in settings such as neurodegeneration, psychiatry, and oncology.

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A Appendix: Methods

A.1 Derivation of the ELBO for the joint log-likelihood

In the main text we defined the joint generative model $p(x, y, z, s) = p(z)p(s)p(x \mid z, s)p(y \mid z)$, and the variational posterior $q(z, s \mid x, y) = q(z \mid x, y)q(s \mid x)$. To define the joint log-likelihood, following [11], we introduce the variational posterior $q(z, s \mid x, y)$ as:

$$\begin{aligned}\log p(x, y) &= \log \int p(x, y, z, s)\, dz\, ds \\ &= \log \int q(z, s \mid x, y) \frac{p(x, y, z, s)}{q(z, s \mid x, y)}\, dz\, ds \\ &= \log \mathbb{E}_{q(z,s|x,y)} \left[\frac{p(x, y, z, s)}{q(z, s \mid x, y)} \right].\end{aligned} \tag{16}$$

Applying Jensen's inequality yields

$$\log p(x, y) \geq \mathbb{E}_{q(z,s|x,y)} \left[\log \frac{p(x, y, z, s)}{q(z, s \mid x, y)} \right]. \tag{17}$$

Using the conditional independence $p(x, y \mid z, s) = p(x \mid z, s)p(y \mid z)$ and the posterior factorization $q(z, s \mid x, y) = q(z \mid x, y)q(s \mid x)$, the variational lower bound of the joint log marginal likelihood is defined as

$$\begin{aligned}\log p(x, y) \geq \quad & \mathbb{E}_{q(z|x,y)q(s|x)}[\log p(x \mid z, s)] + \mathbb{E}_{q(z|x,y)}[\log p(y \mid z)] \\ & - \mathrm{KL}(q(z \mid x, y) \| p(z)) - \mathrm{KL}(q(s \mid x) \| p(s)).\end{aligned} \tag{18}$$

Assuming posterior decoupling $q(z \mid x, y) \approx q(z \mid x) \approx q(z \mid y)$*, the KL term on the common latent can be evenly split as:

$$\mathrm{KL}(q(z \mid x, y) \| p(z)) = \tfrac{1}{2}\mathrm{KL}(q(z \mid x) \| p(z)) + \tfrac{1}{2}\mathrm{KL}(q(z \mid y) \| p(z)). \tag{19}$$

Thus, Eq. (18) can be rewritten as:

$$\begin{aligned}\log p(x, y) \geq\ & \mathbb{E}_{q(z|x)q(s|x)} \log p(x|z, s) + \mathbb{E}_{q(z|y)} \log p(y|z) - \tfrac{1}{2}\mathrm{KL}(q(z|x)||p(z)) \\ & - \tfrac{1}{2}\mathrm{KL}(q(z|y)||p(z)) - \mathrm{KL}(q(s|x)||p(s)),\end{aligned}$$

deriving Eq. (3) of main text.

A.2 Derivation of the ELBO for CASL-VAE

Starting from the approximate joint log-likelihood ELBO in Eq.3,

$$\begin{aligned}\log p(x, y) \geq & \mathbb{E}_{q(z|x)q(s|x)}[\log p(x \mid z, s)] + \mathbb{E}_{q(z|y)}[\log p(y \mid z)] - \tfrac{1}{2}\mathrm{KL}(q(z \mid x) \| p(z)) \\ & - \tfrac{1}{2}\mathrm{KL}(q(z \mid y) \| p(z)) - \mathrm{KL}(q(s \mid x) \| p(s)).\end{aligned} \tag{20}$$

The salient latent variable is hierarchical, $s = (s_{\text{cat}}, s_{\text{cont}})$, with posterior factorization $q(s \mid x) = q(s_{\text{cat}} \mid x)\, q(s_{\text{cont}} \mid s_{\text{cat}}, x)$, and prior factorization $p(s) = p(s_{\text{cat}})\, p(s_{\text{cont}} \mid s_{\text{cat}})$.

The KL divergence for the salient latent is expanded as

$$\begin{aligned}\mathrm{KL}(q(s \mid x) \| p(s)) &= \mathbb{E}_{q(s|x)} \left[\log \frac{q(s \mid x)}{p(s)} \right] \\ &= \mathbb{E}_{q(s_{\text{cat}}|x)q(s_{\text{cont}}|s_{\text{cat}},x)} \left[\log \frac{q(s_{\text{cat}} \mid x)q(s_{\text{cont}} \mid s_{\text{cat}}, x)}{p(s_{\text{cat}})p(s_{\text{cont}} \mid s_{\text{cat}})} \right] \\ &= \mathbb{E}_{q(s_{\text{cat}}|x)} \left[\log \frac{q(s_{\text{cat}} \mid x)}{p(s_{\text{cat}})} \right] + \mathbb{E}_{q(s_{\text{cat}}|x)} \left[\mathbb{E}_{q(s_{\text{cont}}|s_{\text{cat}},x)} \log \frac{q(s_{\text{cont}} \mid s_{\text{cat}}, x)}{p(s_{\text{cont}} \mid s_{\text{cat}})} \right] \\ &= \mathrm{KL}(q(s_{\text{cat}} \mid x) \| p(s_{\text{cat}})) + \mathbb{E}_{q(s_{\text{cat}}|x)} \left[\mathrm{KL}(q(s_{\text{cont}} \mid s_{\text{cat}}, x) \| p(s_{\text{cont}} \mid s_{\text{cat}})) \right].\end{aligned} \tag{21}$$

*Posterior decoupling is enforced via regularization using R_{MMD} and R_{LC} as described in Section 3.6, which encourages $q(z \mid x) \approx q(z \mid y)$.

Substituting Eq. (21) into Eq. (20) gives

$$\log p(x,y) \geq \mathbb{E}_{q(z|x)q(s_{\text{cat}},\, s_{\text{cont}}|x)}[\log p(x \mid z, s_{\text{cat}}, s_{\text{cont}})] + \mathbb{E}_{q(z|y)}[\log p(y \mid z)]$$
$$- \tfrac{1}{2}\mathrm{KL}(q(z \mid x)\|p(z)) - \tfrac{1}{2}\mathrm{KL}(q(z \mid y)\|p(z)) - \mathrm{KL}(q(s_{\text{cat}} \mid x)\|p(s_{\text{cat}})) \quad (22)$$
$$- \mathbb{E}_{q(s_{\text{cat}}|x)}\left[\mathrm{KL}(q(s_{\text{cont}} \mid s_{\text{cat}}, x)\|p(s_{\text{cont}} \mid s_{\text{cat}}))\right]. \quad (23)$$

Finally, separating the domain-specific terms as

$$\mathcal{L}_x = \mathbb{E}_{q(z|x)q(s_{\text{cat}},\, s_{\text{cont}}|x)}[\log p(x \mid z, s_{\text{cat}}, s_{\text{cont}})] - \tfrac{1}{2}\mathrm{KL}(q(z \mid x)\|p(z))$$
$$- \mathrm{KL}(q(s_{\text{cat}} \mid x)\|p(s_{\text{cat}})) - \mathbb{E}_{q(s_{\text{cat}}|x)}\left[\mathrm{KL}(q(s_{\text{cont}} \mid s_{\text{cat}}, x)\|p(s_{\text{cont}} \mid s_{\text{cat}}))\right], \quad (24)$$
$$\mathcal{L}_y = \mathbb{E}_{q(z|y)}[\log p(y \mid z)] - \tfrac{1}{2}\mathrm{KL}(q(z \mid y)\|p(z)), \quad (25)$$

we obtain the ELBO for CASL-VAE, as defined in Eq. (7) (where it is written in terms of parameterized distributions):

$$\mathcal{L}_{ELBO} = \mathcal{L}_x + \mathcal{L}_y.$$

A.3 MMD Regularization for Posterior Decoupling

We use maximum mean discrepancy (MMD) as one of the regularizations to align the common latent distributions $q(z|x)$ and $q(z|y)$, encouraging a domain-invariant common latent representation (Section 3.6).

$$\mathcal{R}_{\text{MMD}} = \text{MMD}^2(q(z|x), q(z|y)) = \mathbb{E}_{z_x, z'_x}[k(z_x, z'_x)] + \mathbb{E}_{z_y, z'_y}[k(z_y, z'_y)] - 2\mathbb{E}_{z_x, z_y}[k(z_x, z_y)]$$

where $k(\cdot,\cdot)$ is a sum of RBF kernels: $k(a,b) = \sum_{\gamma\in\Gamma} \exp(-\gamma\|a-b\|^2)$. We use $\Gamma = \{10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}\}$, corresponding to $\gamma = \frac{1}{2\sigma^2}$.

A.4 Information-Theoretic View of Salient Latent Regularization for the Reference Domain

We analyze how the reference-domain regularizers $\mathcal{R}_{\text{BCE}}$ and $\mathcal{R}_{\text{ref}}$ reduce the mutual information between reference samples $\mathcal{Y}$ and the salient latent $(s_{\text{cat}}, s_{\text{cont}})$.

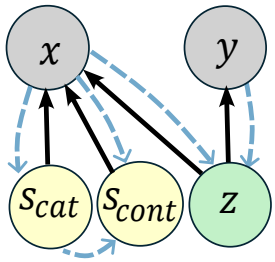


Figure 5: Graphical model showing dependencies between observed data and latent variables. Solid arrows indicate generative paths; dashed arrows indicate inference

Mutual information and variational bound: For random variables X and Z, the mutual information is

$$I(X;Z) = H(Z) - H(Z \mid X), \quad (26)$$

where $H(\cdot)$ denotes Shannon entropy. When Z is represented by any choice of variational posterior $q(z \mid x)$ and prior $p(z)$, a standard variational upper bound [21] is

$$I(X;Z) \;\leq\; \mathbb{E}_{p(x)}[D_{\text{KL}}(q(z \mid x) \,\|\, p(z))]\,. \quad (27)$$

Mutual Information between reference domain and hierarchical salient latent. For reference samples $y \in \mathcal{Y}$, the hierarchical salient latent posterior and prior decomposes as

$$q(s_{\text{cat}}, s_{\text{cont}} \mid y) = q(s_{\text{cat}} \mid y)\, q(s_{\text{cont}} \mid s_{\text{cat}}, y), \qquad p(s_{\text{cat}}, s_{\text{cont}}) = p(s_{\text{cat}})\, p(s_{\text{cont}} \mid s_{\text{cat}}).$$

By the chain rule,

$$I(Y; s_{\text{cat}}, s_{\text{cont}}) = I(Y; s_{\text{cat}}) + I(Y; s_{\text{cont}} \mid s_{\text{cat}}). \quad (28)$$

For the continuous part, applying the variational bound yields

$$I(Y; s_{\text{cont}} \mid s_{\text{cat}}) \leq \mathbb{E}_{p(y)\, q(s_{\text{cat}}|y)}\left[D_{\text{KL}}\left(q(s_{\text{cont}} \mid s_{\text{cat}}, y) \,\|\, p(s_{\text{cont}} \mid s_{\text{cat}})\right)\right]. \quad (29)$$

Thus,

$$I(Y; s_{\text{cat}}, s_{\text{cont}}) \leq I(Y; s_{\text{cat}}) + \mathbb{E}_{p(y)\, q(s_{\text{cat}}|y)}\left[D_{\text{KL}}(q(s_{\text{cont}} \mid s_{\text{cat}}, y) \,\|\, p(s_{\text{cont}} \mid s_{\text{cat}}))\right], \quad (30)$$

which quantifies an upper bound on the information encoded in the salient latent variables about the reference samples.

Effect of regularization terms $\mathcal{R}_{BCE}$ and $\mathcal{R}_{ref}$: For reference domain $\mathcal{Y}$,

1. Minimizing $\mathcal{R}_{BCE}$ pushes $q(s_{\text{cat}} \mid y)$ toward a point mass at the preset reference category ($s'_{\text{cat}} = 1$). This reduces the conditional entropy $H(s_{\text{cat}} \mid Y)$ and the marginal entropy $H(s_{\text{cat}})$, thereby minimizing $I(Y; s_{\text{cat}})$.

2. The KL-divergence regularization $\mathcal{R}_{\text{ref}}$ on continuous salient latent enforces that both the posterior $q(s_{\text{cont}} \mid s_{\text{cat}} = 1, y)$ and the learned prior $p(s_{\text{cont}} \mid s_{\text{cat}} = 1)$ match $\mathcal{N}(0, I)$. By the variational upper bound in Eq. 29, minimizing $\mathcal{R}_{\text{ref}}$ thereby minimizes the upper bound on $I(Y; s_{\text{cont}} \mid s_{\text{cat}})$.

Therefore, $\mathcal{R}_{\text{BCE}}$ and $\mathcal{R}_{\text{ref}}$ jointly minimize an upper bound on $I(Y; s_{\text{cat}}, s_{\text{cont}})$ (Eq. 28), making the salient latent uninformative about the reference domain and forcing it to encode target-specific variation.

A.5 Complementary Roles of the Different Regularization Terms

The regularization terms introduced in Section 3.6 impose complementary inductive biases to disentangle common (shared across reference and target) and salient (target-specific) variation. Latent variable models are generally non-identifiable without strong assumptions [27], and we do not claim formal identifiability guarantees. However, the proposed regularization terms impose constraints that substantially reduce the space of admissible solutions and promote common–salient decomposition in practice. Under these constraints, any solution in which (i) the common latent encodes target-specific variation or (ii) the salient latent encodes reference-domain variation incurs a penalty in the objective, ruling out these classes of degenerate factorizations. Their effects are best understood jointly:

$\mathcal{R}_{\text{MMD}}$ **(domain invariance of common latent)**: aligns the common latent posterior distribution of reference and target domains. For any distribution p, q: $\text{MMD}(p, q) = 0$ iff $p = q$ [17]. Therefore, minimizing this term constrains the common latent z to be domain-invariant, i.e., it discourages encoding of domain-specific information [28]. Consequently, any variation that distinguishes target from reference must be represented in the salient latent space.

$\mathcal{R}_{\text{LC}}$ **(sample identity preservation)**: enforces pairwise consistency of the common latent under cross-domain reconstruction, encouraging z to capture sample-specific, domain-invariant characteristics [45]. However, in isolation it may admit a degenerate solution (e.g. trivially encoding all information in z); therefore, the MMD constraint is important, together, the two terms ensure that z captures identity preserving sample characteristics that exclude target domain effects.

$\mathcal{R}_{\text{BCE}}$ **and** $\mathcal{R}_{\text{ref}}$ **(Salient latent regularization)**: BCE term forces separation between reference and target domain. It assigns all reference samples to a preset salient category (category 1), while forcing target samples to occupy the remaining categories, encouraging salient latents 2,..,M to encode target-specific variation. The $\mathcal{R}_{\text{ref}}$ term enforces posterior collapse of the conditional (continuous) salient latent for the preset category 1. As described in Appendix A.4, these terms jointly minimize an upper bound over the mutual information between the reference domain and salient latent space ensuring that salient latent is uninformative for the reference domain and captures only target-specific variation.

A.6 Implementation Details

A.6.1 Model Architecture

The overall architecture of CASL-VAE is illustrated in Fig. 1a. The common encoder e_{ϕ_z} maps input data to the parameters of the common latent posterior, producing a mean vector and a log-standard-deviation vector (each of dimension 15). Let the input features be D-dimensional. The salient encoder e_{ϕ_s} first projects the input into a $D/4$-dimensional embedding. This embedding is then mapped to M-dimensional logits, which parameterize the discrete salient posterior $q_{\phi_s}(s_{\text{cat}} \mid x)$. The sampled discrete salient variable is concatenated with the $D/4$-dimensional embedding and passed through additional layers to produce the conditional continuous salient posterior mean (with a dimension 2), corresponding to $q_{\phi_s}(s_{\text{cont}} \mid s_{\text{cat}}, x)$.

The common decoder d_{θ_z} reconstructs the shared component from the sampled common latent representation, while the salient decoder d_{θ_s} reconstructs the target-specific component from the concatenated sampled discrete and continuous salient latents. The outputs of the two decoders are added to form the final reconstruction of the input (with dimension D).

Table 4 summarizes the network architectures of the common encoder and decoder e_{ϕ_z} and d_{θ_z}. Table 5 summarizes the network architectures of the salient encoder and decoder e_{ϕ_s} and d_{θ_s}, as well as the learned prior mean embedding μ_ψ.

Table 4: Architecture of common encoder network e_{ϕ_z} and decoder network d_{θ_z}

Layer	Input Size	Bias Term	Leaky ReLU α	Output Size
Common Encoder e_{ϕ_z}				
Linear1 + Leaky-ReLU	D	Yes	0.2	$D/2$
Linear2 + Leaky-ReLU	$D/2$	Yes	0.2	$D/4$
Linear3	$D/4$	Yes	NA	30
(e_{ϕ_z} output split into 15-dim mean μ_z and log-standard-deviation $\log \sigma_z$)				
Common Decoder d_{ϕ_z}				
Linear1 + Leaky-ReLU	15	Yes	0.2	$D/4$
Linear2 + Leaky-ReLU	$D/4$	Yes	0.2	$D/2$
Linear3	$D/2$	Yes	NA	D

Table 5: Architecture of salient encoder network e_{ϕ_s}, decoder network d_{θ_s}, and the learned prior mean embedding network μ_ψ.

Layer	Input Size	Bias Term	Leaky ReLU α	Output Size
Salient Encoder e_{ϕ_s}				
Input embedding $e_{\phi_s}^{\text{embed}}$				
Linear1 + Leaky-ReLU	D	Yes	0.2	$D/2$
Linear2 + Leaky-ReLU	$D/2$	Yes	0.2	$D/4$
Discrete salient component $e_{\phi_s}^{\text{cat}}$				
Linear3 + Leaky-ReLU	$D/4$	Yes	0.2	$D/4$
Linear4	$D/4$	Yes	NA	M
Continuous salient component $e_{\phi_s}^{\text{cont}}$				
Linear5 + Leaky-ReLU	$D/4$	Yes	0.2	$D/4$
Linear6	$D/4$	Yes	NA	2
($e_{\phi_s}^{cat}$ outputs M-dim logits, $e_{\phi_s}^{cont}$ outputs 2-dim mean μ_s)				
Salient prior mean embedding μ_ψ				
Linear1 + Leaky-ReLU	M	Yes	0.2	$D/4$
Linear2	$D/4$	Yes	NA	2
Salient Decoder d_{ϕ_s}				
Projection of discrete salient latent				
Linear1	M	Yes	NA	$D/16$
Projection of continuous salient latent				
Linear2	2	Yes	NA	$D/16$
Concatenate discrete and continuous salient latent and decode				
Linear3 + Leaky-ReLU	$D/8$	Yes	0.2	$D/4$
Linear4 + Leaky-ReLU	$D/4$	Yes	0.2	$D/2$
Linear5	$D/2$	Yes	NA	D

A.6.2 Training Details

The model was trained using the ADAM optimizer [25] with a learning rate of 1×10^{-3}. The optimizer parameters were set to $\beta_1 = 0.5$ and $\beta_2 = 0.999$. For all experiments, we used the following default regularization hyperparameters: $\lambda = 100$, $\eta = 10$, $\zeta = 1$, and $\gamma = 1$. We found the model to be largely robust to variations in these hyperparameters, as demonstrated in Appendix B.4. Among all regularization weights, removing the λ and ζ terms had the most noticeable effect on model performance.

The batch size was set to 128, and training was run for a maximum of 500 epochs. The temperature τ was initialized to 1 and annealed during training, and early stopping was considered only after $\tau < 0.5$. After this point, training was terminated when both losses $\mathcal{L}_x$ and $\mathcal{L}_y$ stabilized and cluster assignments converged.

All experiments were run on a single-node GPU cluster (NVIDIA P100, 12 GB GPU memory, 256 GB system memory). The maximum runtime for the most computationally expensive procedure (clustering stability analysis for selecting M) was approximately 3 hours, with shorter runtimes for all other experiments.

A.6.3 CASL-VAE Algorithm

CASL-VAE training procedure is detailed in Algorithm 1.

Algorithm 1 CASL-VAE training procedure.

Input: Target samples $\mathcal{X}$, reference samples $\mathcal{Y}$; encoders e_{ϕ_z}, e_{ϕ_s}; decoders $d_{\theta_z}, d_{\theta_s}$; batch size B; initial temperature τ_0; minimum temperature $\tau_{\min}$; decay rate r; maximum steps S.
Initialize: $\tau \leftarrow \tau_0$, learnable prior mean embedding parameters ψ
Learned prior: $p(s_{\text{cont}} \mid s_{\text{cat}}) = \mathcal{N}(\mu_\psi(s_{\text{cat}}), I)$
for $step = 1$ **to** S **do**
  if stopping criterion is met **then**
    break
  end if
  if $step \bmod 100 = 0$ **then**
    $\tau \leftarrow \max(\tau_{\min}, \tau_0 \exp(-r \cdot step))$
  end if
  Sample minibatches $\{x_b\}_{b=1}^B \sim \mathcal{X}$ and $\{y_b\}_{b=1}^B \sim \mathcal{Y}$
  Encode common latent:

$$(\mu_z(x_b), \sigma_z(x_b)) = e_{\phi_z}(x_b), \quad z_x \sim \mathcal{N}\big(\mu_z(x_b), \sigma_z^2(x_b) I\big)$$

$$(\mu_z(y_b), \sigma_z(y_b)) = e_{\phi_z}(y_b), \quad z_y \sim \mathcal{N}\big(\mu_z(y_b), \sigma_z^2(y_b) I\big)$$

  Encode salient latents for target samples:

$$\text{proj}_x = e_{\phi_s}^{\text{embed}}(x_b)$$

$$\alpha(x_b) = e_{\phi_s}^{\text{cat}}(\text{proj}_x) \quad \Rightarrow \quad s_{\text{cat},x} \sim \text{Gumbel-Softmax}(\alpha(x_b), \tau)$$

$$\mu_s(s_{\text{cat},x}, x_b) = e_{\phi_s}^{\text{cont}}\big(\text{concat}(s_{\text{cat},x}, \text{proj}_x)\big) \quad \Rightarrow \quad s_{\text{cont},x} \sim \mathcal{N}\big(\mu_s(s_{\text{cat},x}, x_b), I\big)$$

  Fix salient latents for reference samples to preset values:

$$s'_y = \text{concat}(s'_{\text{cat},y} = 1,\ s'_{\text{cont},y} = 0)$$

  Decode reconstructions:

$$\hat{x} = d_{\theta_z}(z_x) + d_{\theta_s}(\text{concat}(s_{\text{cat}_x}, s_{\text{cont}_x}))$$

$$\hat{y} = d_{\theta_z}(z_y) + d_{\theta_s}(s'_y)$$

  Compute ELBO terms $\mathcal{L}_x$, $\mathcal{L}_y$ (Eqs. 8, 9), using the learned prior $p(s_{\text{cont}} \mid s_{\text{cat}})$
  Compute regularizers $\mathcal{R}_{\text{MMD}}$, $\mathcal{R}_{\text{LC}}$, $\mathcal{R}_{\text{BCE}}$, $\mathcal{R}_{\text{ref}}$ (Section 3.6)
  Define objective $\mathcal{J} = -\mathcal{L}_x - \mathcal{L}_y + \lambda \mathcal{R}_{\text{MMD}} + \eta \mathcal{R}_{\text{LC}} + \zeta \mathcal{R}_{\text{BCE}} + \gamma \mathcal{R}_{\text{ref}}$
  Update parameters $\{\phi_z, \phi_s, \theta_z, \theta_s, \psi\}$ using ADAM by minimizing $\mathcal{J}$
end for
Return: trained encoders, decoders, prior mean embedding μ_ψ

B Appendix: Experiments

B.1 Neuroimaging Datasets

B.1.1 Data Preprocessing.

For all neuroimaging experiments, we use data from the iSTAGING consortium [19] that consolidated imaging and clinical data from multiple studies. All T1w MRI scans in the iSTAGING consortium were first corrected for intensity inhomogeneities and skull stripped. Regional brain volumes of 145 ROIs (gray matter: 119 ROIs, white matter: 20 ROIs, ventricles: 6 ROIs) were then extracted [12]

Table 6: Summary of the datasets used in the neuroimaging and model generalization experiments.

Dataset	Age (mean ± SD)	Male (%)	Total N	Diagnostic/group composition
iSTAGING Semi-synthetic cohort construction	63.01 ± 7.65	49	1,979	CN: 1,979
ADNI2/GO Real-world training cohort	72.42 ± 7.27	52	966	CN: 306; MCI: 503; AD: 157
ADNI1 Real-world testing cohort	75.10 ± 6.83	58	815	CN: 227; MCI: 395; AD: 193
Genomics Cell-response experiment	N/A	N/A	3,500	Control: 1,500; Salmonella: 1,000; *H. polygyrus* (day 10): 1,000
Multi-study neuroimaging cohorts OASIS-3, OASIS-4, and PENN	73.82 ± 10.33	43	2,030	CN: 873; MCI: 337; AD: 820
MNIST Toy image experiment	N/A	N/A	2,800	Reference digit 0: 700; Target digits 1/2/3: 700/700/700

from baseline T1w MRI scans and harmonized across sites using ComBat-GAM [33]. Across all experiments, unpaired reference–target samples were used for model training. Following standard procedure, for each train/test split (80%/20%), all samples were first residualized to remove covariate effects of age, sex, and intracranial volume (ICV) using linear regression coefficients estimated from the *training reference* dataset. The residualized features were then standardized with respect to the *training reference* dataset.

B.1.2 Semi-synthetic Data Construction

To have ground-truth disease patterns and severity, we simulated disease effects in real cognitively normal (CN) samples from iSTAGING consortium (Appendix Table 6). Specifically, we randomly split 988 CN (age 45-79) participants into a CN reference ($N = 494$) group and a pseudo-target ($N = 494$) group. The pseudo-target group was then randomly divided into three disease subtypes (S1–S3, Appendix Table 7), each defined by a unique pattern of 14 ROIs (with some overlap across subtypes). For each pseudo-target sample, subtype-specific ROIs were reduced by a rate uniformly sampled between 10–30%, representing the typical atrophy range. To evaluate model robustness to effect size, additional datasets were created with narrower ranges: 0–10% (extremely mild), 10–20% (mild), and 20–30% (moderate). Additionally, to evaluate conditional paired-sample generation, we applied the above simulation to held-out CN samples ($N = 496$) to generate corresponding *paired* pseudo-target samples, creating a semi-synthetic test set with ground-truth pairings.

Table 7: Brain regions included in the semi-synthetic disease patterns used in this study. Each subtype (S1–S3) affects 7 bilateral brain regions (left and right hemispheres; 14 ROIs in total), with partial overlap across subtypes. OP: opercular part.

Brain Region (bilateral)	S1	S2	S3
Amygdala	✓		
Temporal pole	✓	✓	✓
Hippocampus	✓	✓	
Angular gyrus			✓
Entorhinal area	✓		
Frontal operculum		✓	
Inferior temporal gyrus	✓		
Lateral orbital gyrus			✓
Medial frontal cortex		✓	✓
Middle frontal gyrus		✓	
Middle occipital gyrus			✓
OP of inferior frontal gyrus		✓	
Parahippocampal gyrus	✓		
Posterior insula		✓	
Parietal operculum	✓		✓
Supramarginal gyrus			✓

B.1.3 Real Alzheimer's Disease Cohort

Alzheimer's Disease Neuroimaging Initiative (**ADNI**) [5] includes imaging and clinical data from cognitively normal (CN) participants as well as clinically diagnosed mild cognitive impairment (MCI) or Alzheimer's disease (AD) participants. The model was trained on ADNI2/GO participants (306 CN samples, 660 MCI/AD samples) and evaluated on held-out ADNI1 participants (227 CN samples, 588 MCI/AD samples) (see Appendix Table 6). Details on available clinical variables used for downstream analysis can be found in Appendix B.6.2.

B.2 Experiment Procedure

For fair comparison in the semi-synthetic neuroimaging experiments, the same data preprocessing pipeline is used for all baseline methods (K-means, GMM, GMVAE, cVAE, MM-cVAE, SepVAE, SmileGAN; refer Section B.1.1).

CASL-VAE uses a common latent space of dimension $k = 15$, a single discrete salient latent with M categories, and a continuous salient latent of dimension $t = 2$ (Appendix A.6.1). To ensure comparable model capacity across VAE-based methods, we fixed the total latent dimensionality to 18 for all models. Output layers were linear with Gaussian likelihood. The baselines were configured to match this total capacity while respecting their original design:

- **cVAE**: a single unified latent space of dimension 18.
- **MM-cVAE, SepVAE**: common latent dimension 15 and salient latent dimension 3. MM-cVAE and SepVAE were implemented with a similar encoder–decoder architecture as CASL-VAE, but without a hierarchical salient latent structure.
- **GMVAE**: a single discrete mixture component with M categories and a continuous within-component latent dimension 17. To prevent mixture collapse and ensure subtype separation, we used identity covariance and learned component-specific means conditioned on the discrete mixture variable, following [36].

Model configurations: We evaluate clustering under two settings: (i) target-only clustering and (ii) target+reference clustering. K-means and GMM were applied to both settings. GMVAE is an unsupervised deep clustering model and was therefore applied only to target-only clustering. SmileGAN is a semi-supervised deep clustering method that uses a Generative Adversarial Network to identify target subtypes by contrasting against the reference domain; thus it can be used for target-only clustering but not for target+reference clustering. cVAE, MM-cVAE, and SepVAE are generative models trained on both target and reference data, with SepVAE additionally enforcing separation of reference and target samples in its salient latent space using a BCE regularization term in its objective. To assess clustering performance for these models, we apply K-means to their inferred latent representations (unified latent for cVAE, salient latent for MMcVAE and SepVAE).

Training and evaluation: All models were trained using the Adam optimizer [25] with a learning rate of 10^{-3} and a batch size of 128. Models were trained 10 times using repeated hold-out cross-validation on unpaired reference and target samples. In each repetition, 80% of the data was used for training and the remaining 20% of the data was held out for assessing model performance. Reported results correspond to the mean and standard deviation or 95% confidence intervals (computed assuming a Student's t-distribution) across repetitions.

Clustering performance and reconstruction quality were evaluated using the Adjusted Rand Index (ARI) and mean absolute error (MAE), respectively, computed on the held-out validation set for each repetition. For paired-sample generation, we additionally evaluated on a separate semi-synthetic test set containing ground-truth paired reference–target samples. Specifically, for each training repetition, the trained model generated paired reference-domain samples for held-out semi-synthetic target samples, and cosine similarity was computed between the generated and true ground-truth paired reference samples.

GMVAE does not support paired-sample generation because it was implemented for target-only clustering and does not model the reference domain, so it was excluded from this evaluation. For cVAE, paired-samples were generated by encoding target samples into the latent representation and decoding them conditioned on the reference domain label. For SepVAE and MM-cVAE, the common

latent was inferred from the target samples, while the salient latent was fixed to an uninformative vector **0** during decoding. CASL-VAE follows the same procedure, except the uninformative vector is set as $(s'_{\text{cat}} = 1, s'_{\text{cont}} = 0)$ (refer Eq. 12).

B.3 Clustering Stability-Based Selection of the Categorical Salient Dimension M

Clustering stability, measured by the mean pairwise Adjusted Rand Index (ARI) across repeated hold-out cross-validation runs, is used to select the categorical salient dimension M and assess the reproducibility of the resulting partitions. We validate this model-selection criterion on semi-synthetic neuroimaging datasets with three target subtypes and one reference category (ground-truth $M = 4$).

In the mild (10-20%), moderate (20-30%), and typical (10-30%) atrophy regimes, the criterion correctly selects $M = 4$, with high clustering stability (Table 8), indicating reproducible partitions across training runs. These results support stability-based model selection when target-specific variation is sufficiently pronounced. In contrast, the 0-10% atrophy regime exhibits uniformly low clustering stability with no clear optimum, consistent with the limited separability of target-specific variation in this regime (Appendix B.4.1). Such low scores may indicate insufficient evidence for reliable subtype discovery rather than support for any particular value of M.

Table 8: Clustering stability (mean pairwise ARI) across different atrophy regimes and categorical salient dimensions (M). Values are reported as mean $\pm$ standard deviation.

Atrophy rate (%)	$M = 2$	$M = 3$	$M = 4$	$M = 5$
0–10	0.020 ± 0.047	0.081 ± 0.087	0.077 ± 0.047	0.064 ± 0.038
10–20	0.27 ± 0.15	0.45 ± 0.14	$\mathbf{0.74 \pm 0.06}$	0.44 ± 0.08
20–30	0.25 ± 0.12	0.33 ± 0.33	$\mathbf{0.91 \pm 0.06}$	0.73 ± 0.13
10–30	0.25 ± 0.10	0.31 ± 0.28	$\mathbf{0.72 \pm 0.12}$	0.66 ± 0.07

B.4 Ablation and Sensitivity Analyses

B.4.1 Sensitivity to Target-Effect Size and Regularization Hyperparameter Values

To assess model sensitivity to effect size of target-specific variation, we trained CASL-VAE on semi-synthetic unpaired reference-target data where the target samples were simulated with extremely mild (0-10%), mild (10-20%), and moderate (20-30%) atrophy rates. We also assessed robustness of model performance to hyperparameter-selection. For transparency, we also performed this analysis on the dataset simulated with the typical atrophy rate (10-30%).

For each semi-synthetic dataset, we trained CASL-VAE with varying hyper-parameters. Specifically, we changed each parameter from 0% (no regularization) to 200% of the preset hyperparameter value (defined in Appendix A.6.2), while keeping all other parameters fixed to the preset values. For each of these ablations, we trained the model 10 times using repeated hold-out cross-validation analysis described in Appendix B.2, and reported the average model performance on the held-out set per training loop.

Figure 6(a) reports the Adjusted Rand Index (ARI) for target-only clustering, while Figure 6(b) reports ARI for joint (target+reference) clustering. ARI was computed between the predicted cluster assignments and the ground-truth subtype labels in the semi-synthetic experiments.

Table 9: Ablation study of individual regularization terms using semi-synthetic data with typical (10-30%) atrophy rate. We activate one regularization term at a time with corresponding hyperparameter set to the default values. Mean ARI over repeated cross-validation is reported for target-only and joint (reference + target) clustering. **Bold**:best.

R_{MMD}	R_{LC}	R_{BCE}	R_{ref}	ARI↑ (target only)	ARI↑ (target+reference)
–	–	–	–	0.5647	0.3486
✓	–	–	–	0.6063	0.4056
–	✓	–	–	0.5856	0.3098
–	–	✓	–	0.4908	0.6489
–	–	–	✓	0.5855	0.5767
✓	✓	✓	✓	**0.6237**	**0.6969**

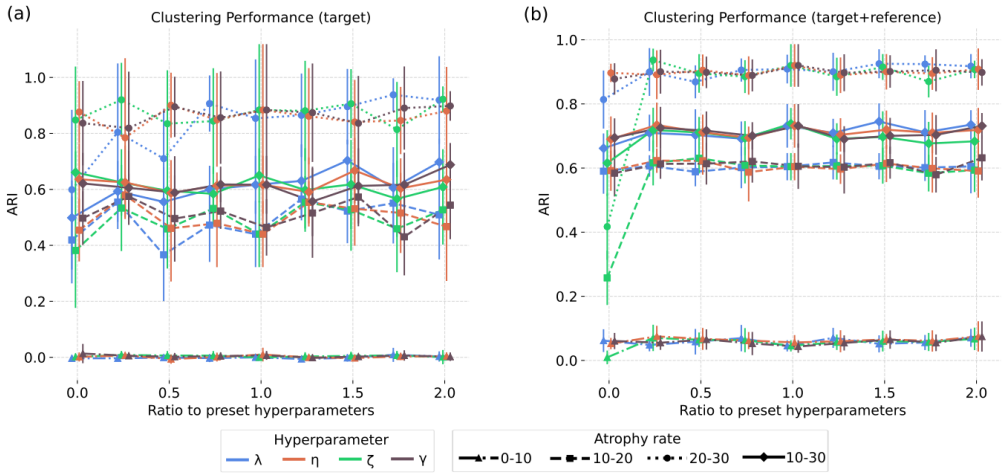


Figure 6: Ablation analysis of CASL-VAE clustering performance under varying regularization strengths and atrophy rates. (a) ARI for target-only clustering and (b) ARI for target+reference clustering, evaluated across 10 repeated hold-out validation splits. Curves show mean ± standard deviation across repetitions. Each regularization weight is scaled relative to its default value, with ratios ranging from 0 to 2 in increments of 0.25. Here, λ controls $\mathcal{R}_{\mathrm{MMD}}$, η controls $\mathcal{R}_{\mathrm{LC}}$, ζ controls $\mathcal{R}_{\mathrm{BCE}}$, and γ controls $\mathcal{R}_{\mathrm{ref}}$.

We observe that CASL-VAE's ability to recover target subtypes degrades rapidly under extremely mild atrophy (0–10%), with ARI (target) approaching zero, indicating near-random cluster assignment. This suggests that when the effect size of target-specific variation is very small, reliable subtype discovery becomes challenging. This behavior is expected in a contrastive setting: when target-specific effects are very small, the target distribution substantially overlaps with the reference distribution, making reliable subtype discovery inherently challenging. As the effect size increases, target-specific variation becomes more distinguishable from the reference domain, and clustering performance improves consistently. A similar trend is observed for target+reference clustering, where stronger target-specific effects lead to clearer separation between reference samples and target subtypes.

Ablation results further highlight the role of individual regularization terms. Each regularizer contributes differently, but performance is strongest when all are combined (Table 9). This is also evident in Figure 6. Removing the MMD regularization on the common latent space leads to a noticeable drop in ARI, indicating degraded domain-invariant representation learning of the common latent. This suggests that, without MMD, target-specific variation may partially leak into the common latent space, weakening the separation between shared and salient factors. Likewise, removing the BCE loss substantially reduces performance in separating reference samples from target subtypes. Importantly, once the regularization terms are active, CASL-VAE exhibits stable performance across a broad range of hyperparameter strengths. Performance degrades primarily when specific regularization terms are removed entirely, demonstrating both the necessity of these components and robustness to precise hyperparameter tuning.

B.4.2 Sensitivity to Latent Dimensionality

We assessed the sensitivity of the model to variations in both the continuous salient dimensionality (t) and the common latent dimensionality (k) using the semi-synthetic dataset with 10–30% atrophy.

Increasing t had minimal impact on subtype recovery (ARI) or reconstruction error (MAE) (Table 10). Although $t = 15$ achieved slightly higher target subtype recovery and lower target reconstruction error, these differences were small relative to the variability across folds. We selected $t = 2$ for direct 2-D visualization of within-subtype variation. More complex datasets exhibiting richer within-subtype heterogeneity may benefit from larger values of t.

Table 10: Varying the continuous salient latent dimension t, while the common latent dimension is fixed at $k = 15$. Performance is reported as mean $\pm$ std across folds.

t	**ARI ↑ (target only)**	**ARI ↑ (reference + target)**	**MAE ↓ (target)**	**MAE ↓ (reference)**
2	0.668 ± 0.087	0.719 ± 0.045	0.607 ± 0.009	$\mathbf{0.622 \pm 0.010}$
5	0.667 ± 0.115	0.726 ± 0.064	0.610 ± 0.011	0.630 ± 0.009
10	0.667 ± 0.066	0.728 ± 0.051	0.610 ± 0.009	0.632 ± 0.015
15	$\mathbf{0.692 \pm 0.057}$	$\mathbf{0.729 \pm 0.047}$	$\mathbf{0.604 \pm 0.006}$	0.627 ± 0.013

Table 11: Varying the common latent dimension k, while the continuous salient latent dimension is fixed at $t = 2$. Performance is reported as mean $\pm$ std across folds.

k	**ARI ↑ (target only)**	**ARI ↑ (reference + target)**	**MAE ↓ (target)**	**MAE ↓ (reference)**
2	0.614 ± 0.113	0.703 ± 0.057	0.705 ± 0.013	0.728 ± 0.012
5	0.626 ± 0.118	0.714 ± 0.061	0.677 ± 0.008	0.696 ± 0.008
10	0.646 ± 0.078	0.712 ± 0.042	0.625 ± 0.007	0.646 ± 0.011
15	$\mathbf{0.668 \pm 0.087}$	$\mathbf{0.719 \pm 0.045}$	$\mathbf{0.607 \pm 0.009}$	0.622 ± 0.010
25	0.593 ± 0.117	0.700 ± 0.061	$\mathbf{0.607 \pm 0.008}$	$\mathbf{0.615 \pm 0.008}$
50	0.615 ± 0.115	0.706 ± 0.056	0.610 ± 0.008	0.624 ± 0.009

Unlike t, the choice of k had a larger effect on reconstruction performance (Table 11). Small values of k substantially increased reconstruction error, suggesting insufficient capacity to capture variation shared across target and reference domains. In contrast, subtype recovery was comparatively robust to the choice of k. Reconstruction performance largely plateaued for $k \geq 15$, although more complex datasets may require larger k.

B.4.3 Controlled Evaluation of Posterior Decoupling

To assess the practical impact of posterior decoupling, we performed a controlled comparison between the paired ELBO in Eq. 18, denoted as L_{paired}, and the decoupled ELBO in Eq. 3, denoted as $L_{decoupled}$. Interpreting this comparison requires some care, however, because the two formulations employ different variational families. The paired formulation uses the joint posterior $q(z|x, y)$, whereas CASL-VAE uses separate posteriors $q(z|x)$ and $q(z|y)$ that are encouraged to be similar. Consequently, differences between L_{paired} and $L_{\text{decoupled}}$ reflect both the posterior-decoupling approximation and the change in variational family. Nevertheless, the resulting ELBO gap provides a practical measure of how closely the decoupled formulation tracks the paired formulation.

We constructed a paired model with the same decoder architecture and latent dimensions as CASL-VAE, but with a common encoder operating on concatenated paired samples. This model was trained using Eq. 18, without the MMD and latent-consistency terms that encourage posterior decoupling in CASL-VAE. CASL-VAE was trained on unpaired samples. Both models were evaluated on held-out paired semi-synthetic samples. The ELBO gap for each training fold was defined as $L_{paired} - L_{decoupled}$ and the relative gap as $100 \times \frac{|L_{paired} - L_{decoupled}|}{|L_{paired}|}$.

As shown in Table 12, the ELBO gap increased with effect size, consistent with posterior decoupling becoming more difficult as target-specific variation strengthens. However, the relative gap remains below 0.6% across all regimes, suggesting that the practical discrepancy between the paired and decoupled objectives is small. We also compared the common latent KL penalties, $KL(q(z|x, y)||p(z)) - (\frac{1}{2}KL(q(z|x)||p(z)) + \frac{1}{2}KL(q(z|y)||p(z)))$, and found that this difference was likewise modest across all settings.

Table 12: Comparison of the paired and decoupled ELBO formulations across simulated atrophy ranges. Results are reported as mean ± std across folds. The ELBO gap is defined as $\mathcal{L}_{\text{paired}} - \mathcal{L}_{\text{decoupled}}$. Statistical significance was assessed using a one-sided Wilcoxon signed-rank test.

Atrophy range (%)	Paired ELBO	Decoupled ELBO	ELBO gap	p-value	Relative gap (%)	Difference in common-latent KL penalty
0–10	-381.98 ± 0.91	-382.38 ± 1.23	0.41 ± 1.25	0.21580	0.31 ± 0.15	2.71 ± 3.44
10–20	-380.97 ± 0.81	-382.57 ± 1.09	1.60 ± 1.15	**0.00683**	0.44 ± 0.27	3.50 ± 3.42
20–30	-379.89 ± 0.99	-382.03 ± 1.07	2.14 ± 1.16	**0.00097**	0.56 ± 0.31	3.03 ± 3.31
10–30	-381.24 ± 0.68	-382.89 ± 0.80	1.65 ± 0.87	**0.00195**	0.44 ± 0.21	3.74 ± 3.49

These results suggest that the decoupled formulation closely matches the paired formulation in the tested semi-synthetic setting, supporting its use for learning the proposed common-salient decomposition.

B.4.4 Robustness to Target Subtype Imbalance

In the 10–30% atrophy regime, we varied the target subtype proportions from balanced 1:1:1 to extremely imbalanced 16:3:1, while fixing $M = 4$. Reconstruction performance on target and reference samples remained stable across imbalance levels, as measured by reconstruction MAE (Table 13). The quality of paired-reference generation was similarly unaffected, with cosine similarity remaining largely unchanged. In contrast, target clustering performance deteriorated with increasing imbalance, with mean ARI decreasing from 0.62 to 0.27. These findings indicate that subtype imbalance primarily affects recovery of subtype structure rather than reconstruction or paired-reference generation.

To further examine this behavior, we increased M to 7 in the moderately imbalanced 5:3:2 setting. The two minority subtypes were recovered in separate categories, whereas the majority subtype was partitioned across the remaining four target categories. These results suggest that increasing M can help preserve minority subtype structure under imbalance, but may also introduce finer subdivisions within majority subtypes. Consequently, the choice of M can influence the trade-off between recovering rare subtypes and avoiding over-partitioning of more prevalent patterns.

Table 13: Sensitivity to target subtype imbalance. Performance is reported as mean ± std across folds.

Target subtype ratio	ARI ↑ (target only)	ARI ↑ (reference + target)	MAE ↓ (target)	MAE ↓ (reference)	Cosine similarity ↑ (paired generation)
1:1:1	0.620 ± 0.113	0.696 ± 0.033	0.607 ± 0.009	0.622 ± 0.009	0.581 ± 0.009
5:3:2	0.582 ± 0.116	0.665 ± 0.044	0.607 ± 0.008	0.625 ± 0.009	0.569 ± 0.011
7:2:1	0.365 ± 0.048	0.609 ± 0.060	0.608 ± 0.009	0.627 ± 0.008	0.568 ± 0.007
16:3:1	0.272 ± 0.036	0.607 ± 0.033	0.609 ± 0.010	0.627 ± 0.008	0.569 ± 0.007

B.4.5 Robustness to Reference-cohort Contamination

We examined contamination of the reference cohort by randomly relabeling target samples as reference in the 10–30% semi-synthetic setting. Such mislabeling increases inter-individual variability within the reference cohort and causes disease-related variation to appear in both domains, reducing the contrast between reference and target populations.

At 10% contamination, mean target-only ARI decreased only slightly from 0.62 to 0.59. Mean target-only ARI further decreased to 0.38 and 0.28 at 30% and 60% contamination, respectively. These findings suggest that CASL-VAE tolerates modest levels of reference-cohort contamination, but subtype recovery deteriorates substantially at higher contamination levels as the reference cohort increasingly contains target-like samples, making target-specific variation harder to isolate.

B.4.6 Model Behavior under Misspecified M

To examine the effect of misspecifying the number of salient categories, we evaluated CASL-VAE with values of M smaller and larger than the ground-truth number of categories in the 10-30% semi-synthetic regime. Because M controls the granularity of the learned target partition, under- and over-specification reveal how the model behaves when the assumed number of categories differs from the underlying subtype structure.

With M=2, CASL-VAE learned a coarse domain level (i.e. reference versus target) separation: 95.4% of CN samples were assigned to the reference category and 88.1% of target samples to the single target-specific salient category. Target samples assigned to the reference category had lower mean atrophy rates than those assigned to the target category (0.16 ± 0.043 versus 0.21 ± 0.057). The resulting target prototype combined regions from all three ground-truth patterns, indicating that an under-specified M merges distinct subtypes into an averaged mode.

With $M = 7$, 92.3% of healthy samples were assigned to the reference category and 93% of target samples occupied the remaining six salient categories. CASL-VAE partitioned each ground-truth subtype into milder and more severe forms: categories 2/6, 4/7, and 5/3 corresponded to S1, S2, and S3, respectively, with mean atrophy rates of 0.23/0.18, 0.25/0.19, and 0.23/0.18. Thus, under-specifying M merged distinct subtype patterns, whereas over-specifying M produced finer partitions that primarily captured within-subtype severity variation rather than inducing categorical collapse. These findings illustrate how the choice of M influences the granularity and interpretation of the learned target-specific structure.

B.5 Supplementary Results from Semi-synthetic Cohort Analysis

Additional results from the semi-synthetic neuroimaging experiment assessing reconstruction quality.

Table 14: Reconstruction performance (MAE) on target domain samples in semi-synthetic data. Values show mean ± std with 95% CI. Lower is better.

Model	MAE ↓	
	10–20%	10–30%
cVAE	0.6355 ± 0.0079 (0.6295, 0.6415)	0.6319 ± 0.0089 (0.6252, 0.6386)
SepVAE	0.6118 ± 0.0076 (0.6061, 0.6175)	0.6101 ± 0.0081 (0.6040, 0.6162)
MM-cVAE	0.6086 ± 0.0074 (0.6030, 0.6142)	0.6086 ± 0.0097 (0.6013, 0.6160)
CASL-VAE (Ours)	**0.6094 ± 0.0116 (0.6006, 0.6181)**	**0.6070 ± 0.0092 (0.6001, 0.6139)**

Table 15: Reconstruction performance (MAE) on CN (reference domain) samples in semi-synthetic data. Values show mean ± std with 95% CI. Lower is better.

Model	MAE ↓	
	10–20%	10–30%
cVAE	0.6394 ± 0.0087 (0.6328, 0.6460)	0.6399 ± 0.0078 (0.6340, 0.6458)
MM-cVAE	0.6225 ± 0.0093 (0.6155, 0.6295)	0.6237 ± 0.0093 (0.6167, 0.6307)
SepVAE	**0.6146 ± 0.0098 (0.6073, 0.6220)**	**0.6151 ± 0.0092 (0.6082, 0.6221)**
CASL-VAE (Ours)	0.6240 ± 0.0121 (0.6149, 0.6331) *	0.6219 ± 0.0095 (0.6147, 0.6291) *

Note: *MAE of CASL-VAE is not significantly different from SepVAE (95% CIs overlap).

B.6 Real-world Disease Cohort Analysis (ADNI)

B.6.1 Diagnostic Composition and Anatomical Characteristics of Learned Subtypes

Table 16 reports the diagnostic composition and MCI progression status of the learned categories in the held-out ADNI1 cohort. Category 3 contained a greater proportion of AD participants and MCI converters than Category 2, indicating an association between category membership and disease burden. However, all three categories contained CN, MCI, and AD participants, and both stable and converting MCI participants were represented across categories. Similar trends were observed

in the training cohort. Thus, while diagnosis and progression status are associated with category membership, they do not fully determine the learned partition.

The learned prototypical patterns (Fig.4) also differed in their spatial distribution of atrophy rather than representing uniformly scaled versions of a common pattern. Category 2 exhibited more diffuse cortical and white-matter alterations, including occipital, parietal, and temporal lobe white matter and the supramarginal and inferior frontal gyri. In contrast, Category 3 showed stronger medial-temporal involvement, with relative preservation of several regions affected in Category 2. These patterns resemble previously reported diffuse cortical and typical AD subtypes, respectively [41]. The distinct regional involvement, together with the substantial diagnostic overlap across categories, suggests that the learned partition captures heterogeneity beyond disease stage alone, although contributions from disease stage or severity cannot be excluded. Determining whether the learned categories reflect distinct biological subtypes, disease stages, or other sources of heterogeneity, including differences in pathology, progression dynamics, and genetic or lifestyle factors, requires further post-hoc validation

Table 16: Diagnostic composition and MCI progression status across CASL-VAE salient categories in the held-out ADNI1 cohort. Category 1 corresponds to the preset reference category. MCI progression status is reported for participants with available labels.

	Category 1	**Category 2**	**Category 3**
Diagnostic composition			
CN	141	60	26
MCI	127	97	171
AD	27	40	126
MCI progression status			
Stable MCI	80	47	54
Converter MCI	45	50	117

B.6.2 Association of Target-specific Salient Variation with Clinical Variables

For each test sample i, CASL-VAE yields an interpretable latent representation $(z_i, s_{\mathrm{cat},i}, s_{\mathrm{cont},i})$. Target-specific variation is quantified by decoding the salient latent and subtracting the reference baseline:

$$\Delta_i = d_{\theta_s}(s_{\mathrm{cat},i}, s_{\mathrm{cont},i}) - d_{\theta_s}(s'_{\mathrm{cat}}, s'_{\mathrm{cont}}). \tag{31}$$

This isolates deviations relative to the reference domain and provides a direct representation of target-specific structure in the input space. To obtain a scalar measure of overall deviation, we compute a deviation index (DI) as the mean absolute value of Δ_i across all dimensions:

$$\mathrm{DI}_i = \frac{1}{D}\sum_{d=1}^{D} |\Delta_{i,d}|\,,$$

where D is the dimensionality of the decoded output. Another way of interpreting this is that DI measures the overall magnitude of an individual's deviation from their paired reference counterpart.

To evaluate whether CASL-VAE captures meaningful disease-related variation in real data, we correlated the deviation index (DI) with clinical variables and biomarkers reflecting AD pathology severity. For each clinical variable, we only selected participants who had the variable available to include in the statistical analysis. The clinical variables used were ADNI-MEM, ADNI-LAN, and ADNI-EF, which are composite scores measuring memory, language, and executive function, respectively (higher values indicate better performance). MMSE is a global cognitive score (higher values indicate better cognition), and ADAS-Cog-13 is another complementary score that measures cognitive performance across 13 tasks, where higher scores indicate worse cognitive performance. APOE E4 carrier status indicates whether the participant carries at least one APOE E4 allele (0 = non-carrier, 1 = carrier), with APOE E4 being a genetic risk factor for AD. We also included cerebrospinal fluid (CSF) biomarkers: CSF-Abeta (amyloid-β 1–42; lower values indicate greater amyloid pathology), CSF-pTau (phosphorylated tau; higher values indicate greater tau pathology), and CSF-Tau (total tau; higher values indicate greater neurodegeneration). For dichotomous variables (APOE E4 carrier), point-biserial correlation was used; Pearson correlation was used for continuous variables.

Table 3 reports the results of this analysis. Higher DI values were associated with lower memory, language, and executive function scores (ADNI-MEM: $r = -0.469$, ADNI-LAN: $r = -0.348$, ADNI-EF: $r = -0.276$), higher overall cognitive impairment (ADAS-Cog-13: $r = 0.473$), lower MMSE scores ($r = -0.354$), and increased likelihood of being an APOE E4 carrier ($r = 0.242$). CSF Abeta showed a negative correlation ($r = -0.254$), while correlations with CSF Tau and pTau were weak and not significant. These results indicate that the salient latent variables in CASL-VAE capture clinically meaningful target-specific variation across samples.

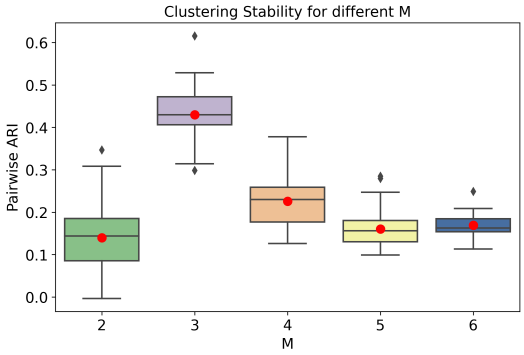


Figure 7: Clustering stability of CASL-VAE on ADNI2/GO training data. Box-plots show pairwise ARI between clusterings from 10 folds of repeated hold-out cross-validation, for $M = 2, \ldots, 6$.

B.7 Results of Generalization Analysis

Appendix Table 6 describes the data used in the model generalization experiments. The code used to preprocess these datasets as well as train the models is available at `https://github.com/SpandanaC-hub/CASLVAE`.

Genomics (cell infection response). We evaluate whether CASL-VAE separates infection states in the unpaired reference-target gene expression dataset. Figure 8 shows PCA projections of the original gene expression measurements along with projections of common latent and salient latent representations. While raw data and common latent representations exhibit overlap between groups, the salient latent representation shows clear separation between infection states and from the reference population. This is reflected in improved clustering performance (Table 17), where CASL-VAE achieves the highest ARI, indicating effective recovery of biologically meaningful subtypes.

Table 17: Clustering performance (mean ARI, 95% CI) on gene expression data. **Bold**: best.

Model	ARI ↑ target-only	ARI ↑ target + reference
SepVAE	0.2157 (0.0760, 0.3553)	0.1699 (0.1296, 0.2102)
MM-cVAE	0.1228 (0.0930, 0.1526)	0.2414 (0.1912, 0.2917)
CASL-VAE	**0.3173 (0.0785, 0.5562)**	**0.7267 (0.6535, 0.7999)**

Table 18: Correlation of clinical and biomarker variables with the overall deviation index (DI) in multi-study (OASIS+PENN) neuroimaging cohort. Variables with $p < 0.001$ are marked accordingly.

Variable	**Correlation with DI**	**p-value**	$p < 0.001$
APOE E4 Carrier	0.115	2.58×10^{-2}	No
MMSE	−0.567	7.30×10^{-53}	Yes
CSF-Abeta	−0.099	5.42×10^{-1}	No
CSF-PTau	0.3368	3.36×10^{-2}	No
CSF-Tau	0.628	1.47×10^{-5}	Yes

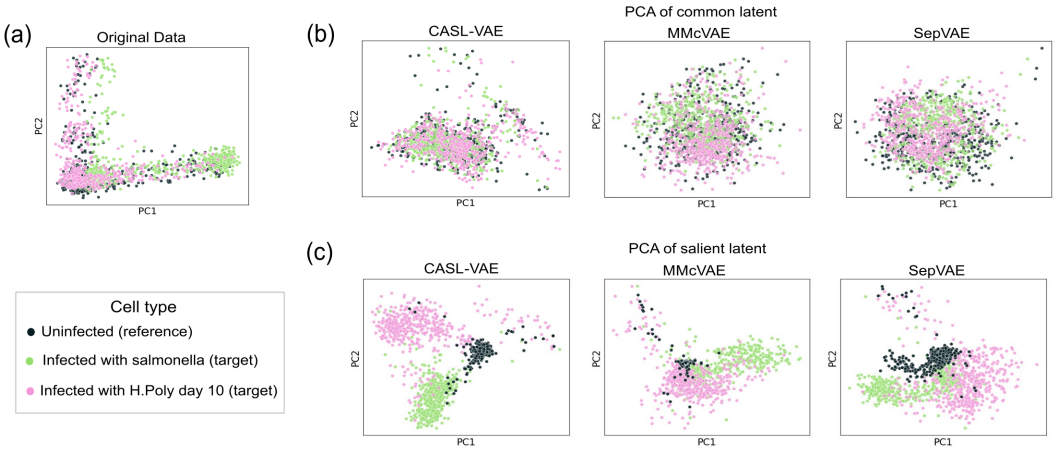


Figure 8: Results from mice intestinal epithelial cell dataset. Plotting the first two principal components from PCA of reference-target a) gene expression measurements, b) common latent representation, and c) salient latent representation.

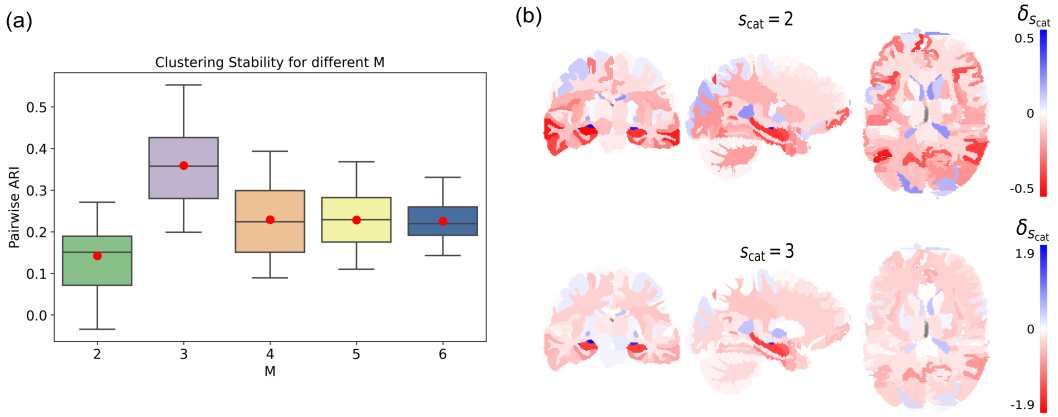


Figure 9: Results from multi-study (OASIS+PENN) neuroimaging cohort analyses. a) Clustering stability of CASL-VAE on multi-study training data. Box-plots show pairwise ARI between clusterings from 10 folds of repeated hold-out cross-validation, for $M = 2, \ldots, 6$. b) Prototypical target patterns $\delta_{s_{\text{cat}}}$ learned in the multi-study cognitively impaired participants relative to CN participants. Negative values (red) indicate volumetric reductions with respect to CN group.

Multi-Study Neuroimaging Cohort. In this analysis we used CN, MCI, AD participant data from the Open Access Series of Imaging Studies (OASIS) and the University of Pennsylvania Memory Center (PENN) cohorts [19]. Based on clustering stability, we select $M = 3$ (Figure 9a), yielding two stable target subtypes. The corresponding patterns (Figure 9b) show atrophy patterns localized to medial temporal regions. Furthermore, the inferred salient neuroanatomical variation is significantly associated with cognitive and biomarker variables (Table 18), supporting the biological plausibility of the learned salient latent representations.

Toy Image Dataset (MNIST-based). We evaluate whether CASL-VAE decomposes common and target-specific salient variation in a synthetic MNIST-based dataset (Figure 10a). To visualize the learned latent factors, we sample from the model priors and decode the resulting samples. The common latent space captures variability associated with digit 0 (Figure 10b), while the salient latent space captures subtype-specific variation corresponding to the superimposed target digits

(Figure 10c,d). These results suggest that CASL-VAE successfully separates shared and salient structure even in non-biological image data.

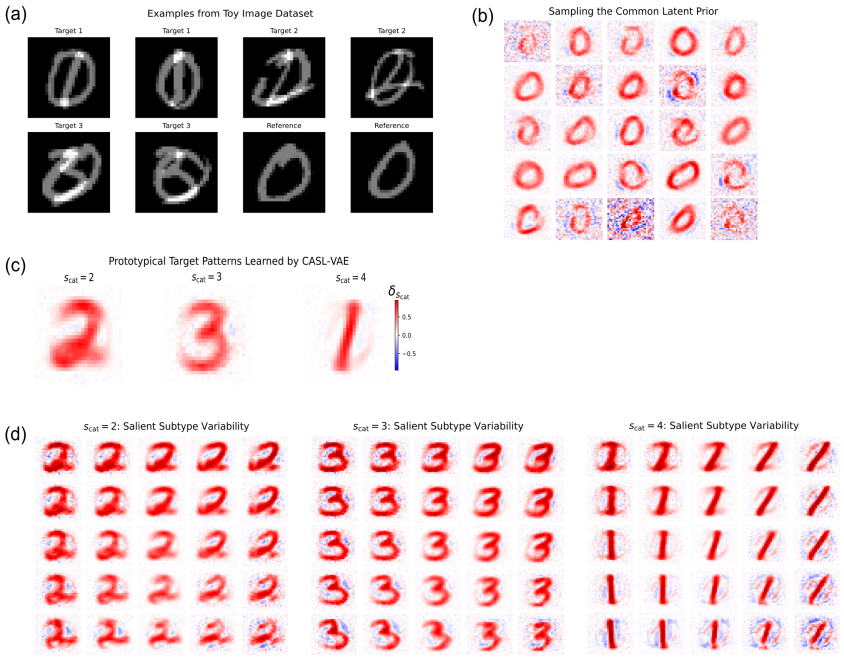


Figure 10: Results from toy image dataset analysis. a) Examples of reference and target images, b) decoding samples from the common latent, c) Prototypical target patterns learnt by CASL-VAE, d) decoding samples from the salient latent.

B.8 Using More Expressive Decoder Architectures

While the additive decoder architecture introduces a tradeoff between interpretability and generative flexibility. Empirically, however, we observe that this structure is sufficient to capture meaningful variation across diverse datasets. Importantly, the additive design can be replaced with more expressive decoders. Following the code we included, replacing the additive decoder with a single decoder architecture that takes all latent variables as input (with all other components unchanged) yields consistent results. In preliminary experiments on ADNI, the learned salient variations are significantly correlated with clinical variables measuring cognitive impairment and Alzheimer's Disease pathology (Table 19).

Table 19: Correlation of ADNI clinical variables with the deviation index (DI). Results from model with a more expressive decoder architecture i.e. a single decoder that jointly decodes common and salient latent representations.

Variable	**Correlation**	$p < 0.001$
ADNI-MEM	-0.435	*
ADNI-LAN	-0.330	*
ADNI-EF	-0.321	*
ADAS-COG-13	0.461	*
APOE E4 Carrier	0.203	*
MMSE	-0.395	*
CSF-Abeta	-0.266	*
CSF-PTau	0.173	
CSF-Tau	0.135	

These results are consistent with those obtained under the additive decoder setting (Table 3), indicating that the framework's behavior is driven primarily by the structured latent decomposition rather than the decoder parameterization.

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